ROTATION INVARIANCE FROM LATTICE SYMMETRY AND REFLECTION POSITIVITY

Abstract. Consider a random distribution with law invariant under translations and satisfying the growth condition in Glimm–Jaffe's regularity axiom. For tensor representations with planar weights of absolute value at most two, we prove that reflection positivity across the coordinate and diagonal hyperplanes implies $\mathrm{O}(d)$-covariance of the correlation functions away from coincidences. The intended application is to the construction of Euclidean quantum fields via a continuum limit of lattice theories: once translation invariance and this growth condition are established, rotation invariance follows from reflection positivity.

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1. Introduction

Fix a number $d \geq 2$, and consider d-dimensional Euclidean spacetime $\mathbb{R}^d$. Our precise setup is described in §1.1. In particular, we have a nuclear space $\mathcal{S} = \mathcal{S}(\mathbb{R}^d; \mathcal{T})$ of test functions on $\mathbb{R}^d$ and a corresponding LCTVS $\mathcal{S}'$ of distributions. Here $\mathcal{T}$ is a finite-dimensional real orthogonal representation of $\mathrm{O}(d)$ whose planar rotation weights lie in $\{-2, -1, 0, 1, 2\}$.* By a "random distribution" is meant a Borel probability measure

$$\mu : \mathrm{Borel}(\mathcal{S}') \to [0, 1] \tag{1}$$

on $\mathcal{S}'$, which we assume possesses well-defined moments $S_N : \mathcal{S}^N \to \mathbb{R}$. Let $\Phi : \mathcal{S} \to \mathbb{C}$ denote the characteristic function

$$\Phi(\chi) = \int_{\mathcal{S}'} e^{iF(\chi)} \, \mathrm{d}\mu(F) \tag{2}$$

of the measure. Let $\mathcal{S}_N \subset \mathcal{S}^N$ denote the set of tuples $\chi = (\chi_1, \ldots, \chi_N)$ such that the test functions $\chi_\bullet$'s have pairwise disjoint supports. The Osterwalder–Schrader covariance axiom is formulated away from coincidences; see [OS73, §2 and §3, E1 and Remark 1].

*The argument allows scalars, vectors, and two-tensors, as well as higher tensors whose planar weights satisfy this restriction.

Let $\mathcal{N}$ denote the set of all hyperplanes $\subset \mathbb{R}^d$ of the form

$$\{x_j = 0\} \quad (0 \le j \le d-1), \qquad \{x_j + x_k = 0\}, \quad \{x_j - x_k = 0\} \tag{3}$$

for $0 \le j < k \le d-1$ in the latter two families.

Theorem A. Suppose that:

(i) There exists a constant $C > 0$ such that

$$\int_{\mathcal{S}'} e^{F(\chi)} \, \mathrm{d}\mu(F) \le e^{C(\|\chi\|_{L^1} + \|\chi\|_{L^2}^2)} \tag{4}$$

for every real $\chi \in \mathcal{S}$.

(ii) The measure is invariant under spacetime translations. That is, for any $T : \mathbb{R}^d \to \mathbb{R}^d$ of the form $T(\mathbf{x}) = \mathbf{x} + \mathbf{a}$ for $\mathbf{a} \in \mathbb{R}^d$,

$$\Phi(T^*\chi) = \Phi(\chi). \tag{5}$$

(iii) The measure is reflection positive across each of the hyperplanes $H \in \mathcal{N}$: for any finite sequence $\chi_0, \chi_1, \cdots \in \mathcal{S}$ of test functions supported on one side of $\mathbb{R}^d \backslash H$, and for any $z_\bullet \in \mathbb{C}$,

$$\sum_{a,b} \bar{z}_a z_b \Phi(\chi_a - \Theta^* \chi_b) \ge 0, \tag{6}$$

where $\Theta : \mathbb{R}^d \to \mathbb{R}^d$ denotes reflection across the hyperplane H.

Then the measure is rotationally-invariant in the sense of Osterwalder–Schrader, meaning $S_N(R^*\chi) = S_N(\chi)$ for all $R \in \mathrm{O}(d)$ and $\chi \in \mathcal{S}_N$. ■□

Here, $R^*\chi = (R^*\chi_1, \ldots, R^*\chi_N)$.

Estimate (i) is the growth condition in Glimm–Jaffe's regularity axiom [GJ87, §6.1, OS1, (6.1.5)].* Sokal [Sok82, (2.27), §2.3] uses Newman's Lee–Yang exponential bound [New75, Theorems 9–10, (3.5) and (4.6)] to control scalar continuum limits. Logarithmic Sobolev inequalities give another route [Led99, Proposition 2.3]; Bauerschmidt and Dagallier [BD24, Theorem 1.1(i)] prove such inequalities for the scalar φ_2^4 and φ_3^4 measures, uniformly in the lattice cutoff when susceptibility is bounded.† Related Gaussian bounds for real plaquette lifts in the finite-volume, two-dimensional Abelian Higgs model appear in Chandra–Chevyrev [CC24, Theorem 4.6 and Corollary 4.8].

Item (iii) is suited to measures constructed as a continuum limit of lattice models: reflections across the hyperplanes $H \in \mathcal{N}$ are symmetries of the lattice $\mathbb{Z}^d$, and reflection positivity passes to a limit of characteristic functions.‡ We expect to apply this in future work.

Remark. The exponential bound eq. (4) yields the well-definedness of the S_N and their basic analytic properties. See §A. In particular, each S_N is a (possibly tensor-valued) tempered distribution on $(\mathbb{R}^d)^N$.

We conflate the tuple $\chi = (\chi_1, \ldots, \chi_N)$ with the function $\chi : (\mathbb{R}^d)^N \to \mathcal{T}^{\otimes N}$ given by $\chi = \chi_1 \boxtimes \cdots \boxtimes \chi_N$.§

Remark. Reflection invariance across a hyperplane follows easily from translation invariance together with reflection positivity across the same hyperplane; see Proposition B.1. Hence the meat of the conclusion of Theorem A is rotation invariance. It remains to consider only $R \in \mathrm{SO}(d)$.

*At $p = 2$ the full axiom also requires local integrability of the two-point function; we use only its growth condition.

†The Lee–Yang bounds involve the two-point form; bounded susceptibility gives L^2 control. The logarithmic Sobolev route uses the L^2 field gradient, with a bounded mean density supplying the L^1 term.

‡Reflection positivity must still be proved for the lattice measure. The symmetry of the lattice identifies the relevant hyperplanes; it does not by itself imply the positivity inequality.

§$(\chi_1 \boxtimes \chi_2 \boxtimes \cdots)(\mathbf{x}, \mathbf{x}', \cdots) = \chi_1(\mathbf{x}) \otimes \chi_2(\mathbf{x}') \otimes \cdots$ for $\mathbf{x}, \mathbf{x}', \cdots \in \mathbb{R}^d$.

Remark. The covariance condition concerns the behavior of the correlation functions outside of the "big diagonal"

$$\Delta = \{(\mathbf{x}_1, \dots, \mathbf{x}_N) \in (\mathbb{R}^d)^N : \mathbf{x}_j = \mathbf{x}_k \text{ for some } j \neq k\}. \tag{7}$$

We think of a point $X \in (\mathbb{R}^d)^N$ as a configuration of N labeled points $\mathbf{x} \in X$. Then $(\mathbb{R}^d)^N \backslash \Delta$ consists of the configurations of distinct points. Covariance in the sense of Osterwalder–Schrader means that the distribution

$$S_N \circ R^* - S_N \tag{8}$$

is supported on Δ for each R.*

One can also ask about the rotation-invariance of the *measure* μ. Under the bound eq. (4), this would be equivalent to $S_N \circ R^* = S_N$ holding on all of $(\mathbb{R}^d)^N$. The theorem above allows the possibility of a non-covariant "contact term" supported on Δ. This qualification is necessary: the hypotheses permit noncovariant contact terms, as the white-noise example in §10 shows. That section also shows that the weight cutoff cannot be enlarged uniformly from two to three. The off-diagonal covariance asserted in Theorem A is the covariance condition in [OS73, §3, E1 and Remark 1]. With the remaining hypotheses, reconstruction is given by [OS75, §IV.1, Theorem E' (or $E'' \to R'$].

1.1. **Setup.** All vector spaces are over the real numbers unless stated otherwise. Abbreviate

$$t = x_0, \quad x = x_1, \quad y = (x_2, \dots, x_{d-1}) \in \mathbb{R}^{d-2}, \quad \mathbf{x} = (t, x, y). \tag{9}$$

The space $\mathcal{S} = \mathcal{S}(\mathbb{R}^d; \mathcal{T})$ is the Fréchet space of real Schwartz tests, and $\mathcal{S}'$ is its continuous dual with the weak-$*$ topology. The Euclidean inner product on $\mathcal{T}$ fixes the pointwise norm and hence the L^1 and L^2 norms. For an isometry $T(\mathbf{x}) = g\mathbf{x} + \mathbf{a}$, our pullback convention is

$$(T^*\chi)(\mathbf{x}) = \rho(g)\chi(T\mathbf{x}), \qquad (T_1T_2)^* = T_2^*T_1^*. \tag{10}$$

Here $\rho(g)$ is the pullback action on the tensor indices; thus $\rho(g_1g_2) = \rho(g_2)\rho(g_1)$. It is an isometry for real g.

Write $\tau_a\chi(\mathbf{x}) = \chi(\mathbf{x} - a)$, so that τ_a moves the support by a and is the pullback of translation by $-a$. On cylinder observables put

$$U(a)F(\chi) = F(\tau_a\chi). \tag{11}$$

All reflection-positivity statements concern real tests supported on one chosen side of the hyperplane. Positivity on the other side follows from reflection invariance. By Proposition B.1, every reflection in the hypotheses is a symmetry of the law. Coordinate reflections and the reflections $x_j = x_k$ generate the signed coordinate permutations. In particular, the quarter turns used below are symmetries; this is a consequence of the hypotheses, not a separate assumption.

We use

$$S_N(\chi_1, \dots, \chi_N) = \int F(\chi_1) \cdots F(\chi_N)\, \mathrm{d}\mu(F), \qquad S_0 = 1. \tag{12}$$

The regularity bound gives, with a constant K depending only on C,

$$|S_N(\chi_1, \dots, \chi_N)| \leq (K\sqrt{N})^N \prod_{j=1}^{N} (\|\chi_j\|_1 + \|\chi_j\|_2). \tag{13}$$

This estimate and the distributional interpretation are proved in §A. All subsequent products of fields lie in every finite $L^p(\mu)$.

The first moment need not vanish. Translation invariance gives $S_1(\chi) = \langle m, \int \chi \rangle$ for a fixed $m \in \mathcal{T}$. † Each required quarter turn fixes m. Since the only planar weight in $\{-2, -1, 0, 1, 2\}$ fixed

*Near a configuration of distinct points, choose pairwise disjoint neighborhoods of its entries. Products of compactly supported tests in these neighborhoods determine a distribution there by the Schwartz kernel theorem. A partition of unity then gives the equivalence between vanishing on such products and support on Δ.

†A translation-invariant distribution is constant: convolution with a smooth test gives a smooth translation-invariant function, and approximation by a unit-mass mollifier gives the assertion.

by a quarter turn is zero, m is fixed by every rotation in that plane. Thus S_1 is already covariant; the same argument uses only the four planar mirrors of Theorem B.

1.2. **Contribution and relation to earlier work.** The new result is Theorem A: translation invariance, the exponential L^1–L^2 bound (4), and reflection positivity across the specified coordinate and diagonal hyperplanes imply covariance away from coincidences for an arbitrary Borel law. This holds in every dimension $d \geq 2$ for every coefficient representation in the stated weight range. The hypotheses concern the Euclidean law itself; no field equation, interaction, or spectrum condition is prescribed.

The proof develops Streater's method of deriving covariance from the spectrum condition and field bounds [Str72, Theorem, p. 119]. Spectral analyticity first restricts the possible transformation weights; field bounds and positivity then exclude the unwanted weights. In $P(\phi)_2$, Streater assumes the forward-cone spectrum condition, local commutativity, and cutoff-uniform field and canonical-momentum bounds. More specifically, the Cauchy–Schwarz comparison giving convexity (132) is the argument of his Lemma 2 [Str72, pp. 113–114], applied here to the continued reflection form. His Lemma 3 [Str72, p. 114] combines that comparison with a linear degree bound and parity; see also Osipov [Osi77, p. 115]. Our comparison with the law rotated through $\pi/4$ in §8 is a different step, which appears to be new. For already covariant Fermi fields, Mardin and Streater [MS76, (4)–(6), pp. 82–83] compare growth under boosts with bounds from the equal-time anticommutation relations to restrict the spinor rank.

The finite-covariance theorem of Bros, Epstein, and Glaser [BEG67, Theorem 1] is a related analytic result: extended-tube analyticity and tempered Laplace-transform bounds imply that only finitely many Lorentz transformation types occur. We do not apply that theorem; our finite angular expansion follows directly from the one-variable continuation and growth estimate below.

Coordinate and diagonal reflection positivity were already available simultaneously in Sokal's lattice construction [Sok82, pp. 328–330]. His coordinate two-point spectral representation is (2.37), and Remark 2 on p. 330 refers to a diagonal analogue. He proposes combining diagonal reflection positivity with Streater's method in §5(2), pp. 374–375, but leaves the required relativistic two-point energy bound conjectural.* The operator inequality $H \geq |P|$ on arbitrary polynomial vectors required here is not established in that published discussion. Here H and P generate time and spatial translations in the Osterwalder–Schrader Hilbert space. We prove it in §4 by expressing the same translated reflection pairing in the coordinate and diagonal Hilbert spaces, and derive from it the complex translation domains used below. This derivation from a finite family of mirrors, without a model Hamiltonian or continuous rotation covariance, appears to be new.

For $P(\phi)_2$ on a spatial circle, Jaffe, Jäkel, and Martinez [JJM13, Proposition VI.1 and (VI.4)–(VI.5)] use temporal and spatial reflection-positive quantizations, together with their Feynman–Kac formula, to prove $H + vP \geq 0$ for $|v| < 1$. That model-specific argument does not give the spectral-measure comparison for arbitrary polynomial vectors used here.

The reflected Hilbert space and positive-time semigroup are the Osterwalder–Schrader construction [OS73, §4.1, (4.3)–(4.9)]. The propagation of positive derivative matrices in Theorem D is Glaser's Taylor-expansion argument [Gla69, §3, p. 14]; his positive-kernel continuation theorem [Gla74, Theorem A] gives the principle behind Lemma 5.1. The slab-source argument adapts the dyadic reflections of Seiler–Simon [SS76, Theorems 4.3 and 7.1] to (4).

The subsequent localization gives the insertion estimate (96), where the insertion is localized in (t, x) and the y variables remain smeared. This estimate under the present abstract hypotheses seems to be new in this generality. It uses $H \geq |P|$ to justify complex translations and the regularity bound to control the insertion; only two coordinates are localized, so the resulting angular Fourier bound is independent of d. Section 3 explains the proof before the detailed arguments.

*The derivative estimate (A.31), p. 379, is a different conclusion; Sokal refers its proof to an unpublished manuscript.

Even for a centered Gaussian law, (4) requires the covariance form to be bounded on L^2: apply the estimate to $u\chi$ and let $u \to \infty$. We claim neither minimality of the assumptions nor a uniform weakening of all earlier hypotheses. The examples in §10 show why the conclusion allows contact terms and why the weight cutoff two cannot be enlarged uniformly.

The principal steps are summarized in Figure I.

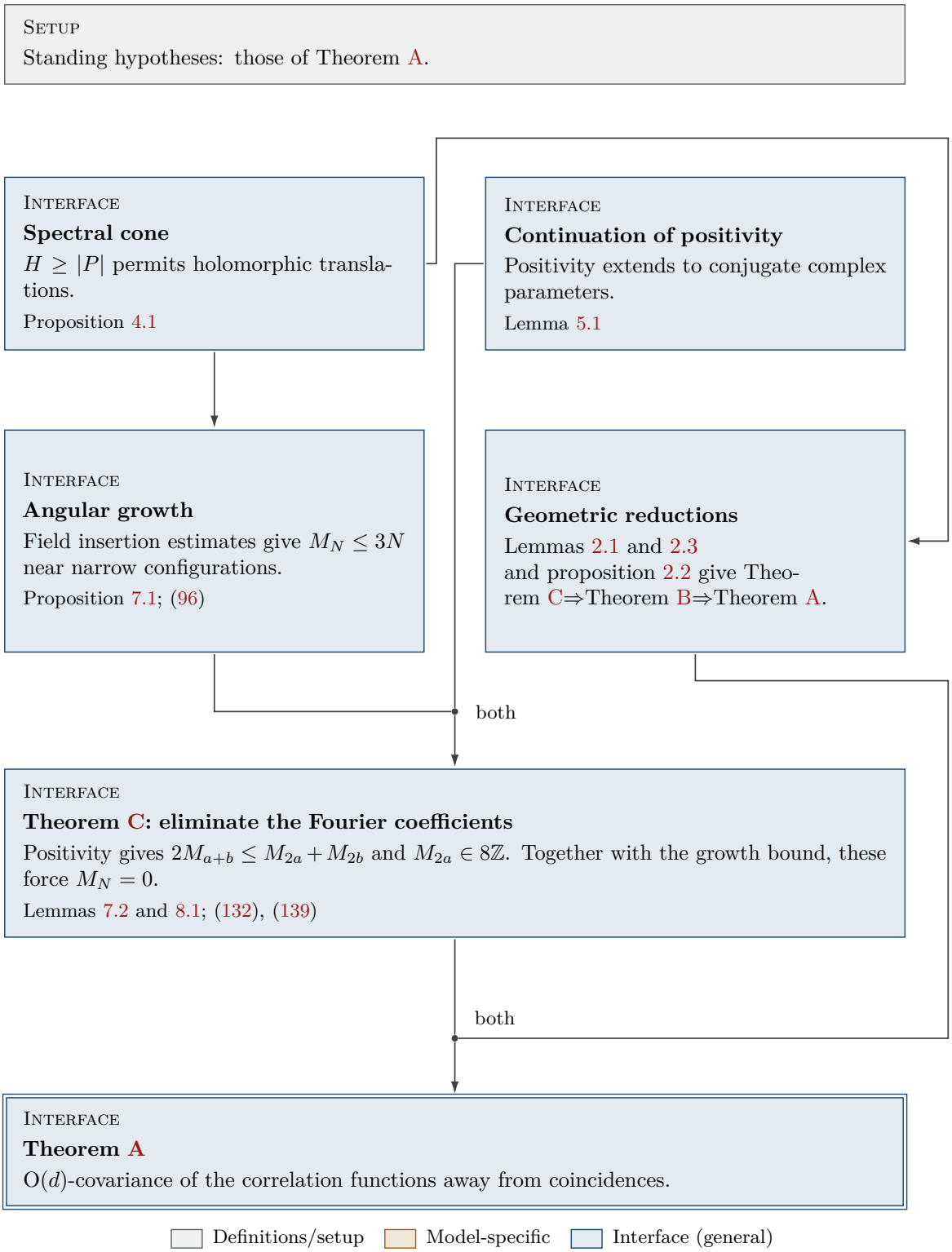


FIGURE I. Principal steps in the proof. Here M_N is the largest absolute angular Fourier frequency at order N near narrow configurations. Arrows record inputs used under the standing hypotheses; both branches at each junction are required. The geometric reductions use analyticity under translations of separated groups.

2. Reductions

For each pair $j < k$, consider the subgroup $G \subset \mathrm{SO}(d)$ consisting of orthogonal transformations R that mix the two coordinates x_j, x_k among themselves and leave the other coordinates fixed. Because $\mathrm{SO}(d)$ is generated by the union of such subgroups, in order to prove full rotation covariance it suffices to prove covariance for transformations lying in each individual G, that is

$$S_N(\chi) = S_N(R^*\chi) \tag{14}$$

for all $N \in \mathbb{N}$, $R \in G$, and $\chi \in \mathcal{S}_N$. The hypotheses of the theorem are invariant under a permutation of the coordinates, so it suffices to consider the group

$$G = \left\{ R(\theta) = \begin{bmatrix} r(\theta) & 0 \\ 0 & I_{d-2} \end{bmatrix} : \theta \in \mathbb{R} \right\} \cong \mathrm{SO}(2), \qquad r(\theta) = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix} \tag{15}$$

of orthogonal transformations of the tx-plane.

For each N-tuple $\chi = (\chi_1, \dots, \chi_N) \in \mathcal{S}_N$ and $\theta \in \mathbb{R}$, define

$$D_N[\chi](\theta) = S_N(R(\theta)^*\chi) - S_N(\chi) \tag{16}$$

to measure the failure of S_N to be rotation-invariant. Because S_N is a distribution, $D_N[\chi] \in C^\infty(\mathbb{S}^1_\theta)$.* Note that $D_N[\chi](0) = 0$. We want to show that

$$D_N[\chi](\theta) = 0 \tag{17}$$

for all θ.

Let $\Delta_2 \subset (\mathbb{R}^d)^N$ be the set of configurations X of points $\mathbf{x} = (t, x, y)$ such that the $(t, x) \in \mathbb{R}^2$ are not all distinct. In other words, if we let $\Pi : \mathbb{R}^d \to \mathbb{R}^2$ denote projection onto the first two coordinates, then $X \in (\mathbb{R}^d)^N$ lies outside Δ_2 if and only if all points $\mathbf{x} \in X$ have distinct projections $\Pi\mathbf{x}$. We prove:

Theorem B. Let $\mu : \mathrm{Borel}(\mathcal{S}') \to [0, 1]$ denote a Borel probability measure on $\mathcal{S}'$. Denote its characteristic function by Φ. Suppose that μ satisfies the following three properties:

(i) There exists a constant $C > 0$ such that

$$\int_{\mathcal{S}'} e^{F(\chi)} \, \mathrm{d}\mu(F) \le e^{C(\|\chi\|_{L^1} + \|\chi\|_{L^2}^2)} \tag{18}$$

for every real $\chi \in \mathcal{S}$.

(ii) The measure is invariant under spacetime translations. That is, for any $T : \mathbb{R}^d \to \mathbb{R}^d$ of the form $T(\mathbf{x}) = \mathbf{x} + \mathbf{a}$ for $\mathbf{a} \in \mathbb{R}^d$,

$$\Phi(T^*\chi) = \Phi(\chi). \tag{19}$$

(iii) The measure is reflection positive across the four hyperplanes

$$\{t = 0\}, \quad \{x = 0\}, \quad \{t + x = 0\}, \quad \{t - x = 0\}. \tag{20}$$

Then $D_N(\theta)$ vanishes on $(\mathbb{R}^d)^N \backslash \Delta_2$ for each $\theta \in \mathbb{R}$. ■□

The conclusion of this theorem is not full Osterwalder–Schrader covariance, because it leaves open the possibility that D_N be nonzero on $\Delta_2 \supsetneq \Delta$, rather than only Δ. In order to deduce full Osterwalder–Schrader covariance (under the hypotheses of Theorem A), an argument is required. The argument exploits some real-analyticity of $D_N[X](\theta)$'s dependence on X to deduce vanishing for $X \in \Delta_2 \backslash \Delta$ from vanishing for $X \notin \Delta_2$. The analytic dependence we use is under translations of the points on one side of a separating coordinate hyperplane, with the other points fixed. This is stated for smeared moments in Lemma 2.1. Analyticity along such translations propagates the

*The notation $D_N[X](\theta)$ below is distributional shorthand: to say that it vanishes near X means that the distribution $D_N(\theta)$ annihilates every test supported in some neighborhood of X. No pointwise evaluation at X is assumed.

vanishing of D_N from a neighborhood of one configuration to neighborhoods of the configurations reached by those translations, as follows.

To make this more precise, we define for each $\theta \in \mathbb{R}$ an equivalence relation $\sim_\theta$ on $(\mathbb{R}^d)^N \backslash \Delta$ as follows. Suppose $X_0, X_1 \in (\mathbb{R}^d)^N \backslash \Delta$ are two configurations for which there exists a one-parameter group of translations $\mathbb{R} \ni s \mapsto T(s)$ of $(\mathbb{R}^d)^N$ with the following properties:

- $T(1)X_0 = X_1$
- for $t \in [0,1]$, $T(t)X_0$ stays inside of $(\mathbb{R}^d)^N \backslash \Delta$,
- for some open neighborhood $O \ni X_0$, if $\chi \in \mathcal{S}_N$ satisfies $\operatorname{supp} \chi \subset O$, then the function

$$s \mapsto D_N[T(-s)^* \chi](\theta) \tag{21}$$

  is real-analytic on some neighborhood of $[0,1]$.

Then we write $X_0 \sim_\theta X_1$. This defines a reflexive symmetric relation, which we extend to an equivalence relation by taking the transitive closure. Then, if $X_0 \sim_\theta X_1$ and $D_N[X](\theta)$ vanishes near $X = X_0$, it also vanishes near $X = X_1$.

Given a configuration $X_0 \in (\mathbb{R}^d)^N \backslash \Delta$, let $H \subset \mathbb{R}^d$ denote an affine hyperplane parallel to the coordinate axes and containing none of the points $\mathbf{x} \in X_0$. Then, H divides the points in X_0 into two groups. Let $\mathbf{n}$ denote a unit vector normal to H. Use that vector to designate one side of H as the forward direction. The following lemma states that $S_N(X)$ is real analytic at $X = X_0$ in any direction moving only the points in the group on the forward side of the hyperplane. More precisely, letting H_0 be the hyperplane parallel to H and passing through the origin:

Lemma 2.1. Fix X_0 as above. For $\delta\mathbf{x} \in \mathbb{R}^d$, write $\tau_{\delta\mathbf{x}}(\mathbf{x}) = \mathbf{x} + \delta\mathbf{x}$, and let $T_\bullet = T_\bullet[H] : (\mathbb{R}^d)^N \to (\mathbb{R}^d)^N$ denote the translation that acts via $\tau_\bullet$ on the jth $\mathbb{R}^d$ factor of $(\mathbb{R}^d)^N$ if the jth point $\in X_0$ lies on the designated side of H (and leaves the other factors fixed.) Then, under the hypotheses of Theorem A, there exists an open neighborhood $O \ni X_0$ such that if $\chi \in \mathcal{S}_N$ is supported in O, then the function

$$(-\varepsilon, \infty) \times H_0 \ni (s, \mathbf{v}) \mapsto S_N(T^*_{-s\mathbf{n}-\mathbf{v}}\chi) \tag{22}$$

is real analytic on $(-\varepsilon, \infty) \times H_0$ for some $\varepsilon > 0$. Under the hypotheses of only Theorem B, the same conclusion holds if the hyperplane is one of constant t or x and $\mathbf{v}$ stays in the tx-plane. ■

The lemma is proved in §6.1. Given the lemma above:

Proof of Theorem A (from Theorem B). Assume the hypotheses of Theorem A. Then Theorem B applies, so $D_N[X](\theta)$ vanishes for $X \in (\mathbb{R}^d)^N \backslash \Delta_2$. Consider a configuration $X \in \Delta_2 \backslash \Delta$. We prove that there exists some $X_0 \in (\mathbb{R}^d)^N \backslash \Delta_2$ such that

$$X \sim_\theta X_0 \tag{23}$$

for all $\theta \in \mathbb{R}$. In light of the discussion above, this will complete the proof: the vanishing of D_N near X_0 implies the vanishing near X. The move is to separate two points with equal planar projection by a hyperplane in a y coordinate and translate one side in time. That hyperplane is fixed by $R(\theta)$, so the two terms of D_N can both be varied analytically using the same separating hyperplane.

For each $\# = 0, \dots, 1 + N(N-1)/2$, let $\Delta_2^\#$ denote the set of configurations $X \in (\mathbb{R}^d)^N$ such that either $X \in \Delta$ or there exists a set of at least $\#$ pairs of distinct $\mathbf{x}, \mathbf{x}' \in X$ such that $\Pi\mathbf{x} = \Pi\mathbf{x}'$. Then

$$(\mathbb{R}^d)^N = \Delta_2^0 \supseteq \Delta_2 = \Delta_2^1 \supseteq \cdots \supseteq \Delta_2^{1+N(N-1)/2} = \Delta. \tag{24}$$

Given $X \in (\mathbb{R}^d)^N \backslash \Delta$, let $\#$ be the smallest nonnegative integer such that $X \in \Delta_2^\# \backslash \Delta_2^{\#+1}$. If $\# = 0$, then $X \in (\mathbb{R}^d)^N \backslash \Delta_2$ already. If instead $\# \geq 1$, let $\mathbf{x}, \mathbf{x}' \in X$ denote two points in the configuration whose first two coordinates agree:

$$\Pi\mathbf{x} = \Pi\mathbf{x}'. \tag{25}$$

Because $X \notin \Delta$, the points $\mathbf{x}, \mathbf{x}'$ are distinct, so they must be distinguished by one of the remaining cartesian coordinates $y_1, \dots, y_{d-2}$. Now pick a hyperplane $H = \{y_\bullet = a\}$ not containing any of the points in X and separating $\mathbf{x}, \mathbf{x}'$.

Above, we defined $T_\bullet$. Redefine this with the coordinate $y_\bullet$ in place of t, using the hyperplane H to define the two groups of points. Thus, for each $\delta t, \delta x \in \mathbb{R}$, we have a translation $T_{\delta t, \delta x}$ (that does nothing to the y coordinates),

$$T_{\delta t, \delta x} : (t, x, y) \mapsto (t + \delta t, x + \delta x, y). \tag{26}$$

Now note that if δt is in some small punctured interval $(-\varepsilon, \varepsilon)\backslash\{0\}$, then

$$T_{\delta t, 0} X \in (\mathbb{R}^d)^N \backslash \Delta_2^{\#} \tag{27}$$

because it removes the coincidence between $\mathbf{x}, \mathbf{x}'$ by changing the time coordinate of one but not the other (and as long as ε is sufficiently small, no more coincidences are created). We will show that

$$X \sim_\theta T_{\delta t, 0} X. \tag{28}$$

If we have already proved the claim for smaller values of $\#$, then $T_{\delta t,0} X \sim_\theta X_0$ for some $X_0 \in (\mathbb{R}^d)^N \backslash \Delta_2$, hence $X \sim_\theta X_0$.

In terms of S_N,

$$\begin{aligned} D_N[T^*_{-\delta t, 0}\chi](\theta) &= S_N[R(\theta)^* T^*_{-\delta t, 0}\chi] - S_N[T^*_{-\delta t, 0}\chi] \\ &= S_N[T^*_{-(\cos\theta)\delta t, -(\sin\theta)\delta t} R(\theta)^* \chi] - S_N[T^*_{-\delta t, 0}\chi]. \end{aligned} \tag{29}$$

Applying Lemma 2.1 with $y_\bullet$ in place of t, we conclude that the right-hand side above depends real analytically on δt near $\delta t = 0$, as long as χ is supported sufficiently close to the configuration X. Hence $X \sim_\theta T_{\delta t, 0} X$. □

The remaining reduction makes the successive planar displacements lie in a common cone of opening less than $\pi/2$. This is the geometry that will permit continuation in every complex rotation angle.

Say that a configuration $X \in (\mathbb{R}^d)^N \backslash \Delta_2$ is *narrow* if the points in X can be listed in some order $\mathbf{x}_1, \dots, \mathbf{x}_N$ such that, for some unit vector $\mathbf{n} \in \mathbb{R}^2$, the differences $\Pi(\mathbf{x}_1 - \mathbf{x}_2), \cdots, \Pi(\mathbf{x}_{N-1} - \mathbf{x}_N) \in \mathbb{R}^2$ all make angle $< \pi/4$ with $\mathbf{n}$. That is,

$$\frac{\mathbf{n} \cdot \Pi(\mathbf{x}_n - \mathbf{x}_{n+1})}{|\Pi(\mathbf{x}_n - \mathbf{x}_{n+1})|} > \frac{1}{\sqrt{2}} \tag{30}$$

for $n = 1, \dots, N-1$.

Proposition 2.2. Under the hypotheses of Theorem B, for any $X \in (\mathbb{R}^d)^N \backslash \Delta_2$, there exists a $\delta > 0$ and narrow X_0 such that $X \sim_\theta X_0$ whenever $-\delta < \theta < \delta$. ■

Proof. First, we show that every $X \in (\mathbb{R}^d)^N \backslash \Delta_2$ is equivalent to a configuration in which all points have a distinct time coordinate. Indeed, proceed via induction on the number of pairs of points sharing a time coordinate. If two times agree, their x coordinates differ. A line $x = a$ separating them and avoiding the configuration divides the points into two groups. Move one group by a sufficiently small amount in the time direction. At least one time coincidence disappears and no new one is created. The two groups remain strictly separated in the x coordinate. By Lemma 2.1, the moment is analytic along this move. For $|\theta|$ small, the rotated groups remain strictly separated in the x coordinate throughout the move, so the other term in D_N is analytic as well. Thus this move is allowed in $\sim_\theta$.

Now suppose that $X \in (\mathbb{R}^d)^N \backslash \Delta_2$ contains points with distinct time coordinates. Then by a series of $T_{\delta t, 0}$ for different hyperplanes, we can stretch X in the vertical direction until the result is a narrow configuration: cut between successive times and move the later group forward until each successive time difference exceeds the absolute difference of the corresponding x coordinates. Every move preserves strict time ordering. Each path is compact, so the same is true after rotation

through sufficiently small $|\theta|$. Apply Lemma 2.1 to both terms of D_N along each path, and take the smallest of the finitely many allowed angle bounds. This gives the claimed δ. The same choices work for configurations in a sufficiently small neighborhood of X. □

We now state:

Theorem C. Under the hypotheses of Theorem B, $D_N[X](\theta)$ vanishes on the set of narrow configurations X. ■□

Given the theorem above:

Proof of Theorem B from Theorem C. Given the hypotheses of the theorem, Theorem C says that $D_N[X](\theta)$ vanishes whenever X is narrow. By Proposition 2.2, we conclude that given any $X \in (\mathbb{R}^d)^N \backslash \Delta_2$, there exists an open neighborhood $O \ni X$ and an interval about $\theta = 0$ such that $D_N[\chi](\theta) = 0$ whenever $\chi \in C_c^\infty(O)$.
Lemma 2.3 completes the proof. □

Given $X \in (\mathbb{R}^d)^N$, let $\mathcal{O}(X) \subset (\mathbb{R}^d)^N$ denote the set of configurations arising by applying to X a common rotation $R(\theta)$ to each point in the configuration. That is, $\mathcal{O}(X)$ is X's orbit under the group of rotations of the tx-plane.

Lemma 2.3. In order to show that $D_N(\theta)$ vanishes near some configuration $X \in (\mathbb{R}^d)^N \backslash \Delta$ for each $\theta \in \mathbb{R}$, it suffices to show that for each $X' \in \mathcal{O}(X)$ there exists an open neighborhood $O \ni X'$ with the following property: there exists a $\delta = \delta(O) > 0$ such that $D_N[\chi](\theta) = 0$ whenever $-\delta < \theta < \delta$ and $\chi \in \mathcal{S}_N$ is supported in O. ■

Proof. Suppose that the desired neighborhoods exist. Because $\mathcal{O}(X)$ is compact, a finite collection of the given neighborhoods covers it. Shrink $O_0 \ni X$ and take the minimum of their angle bounds so that every rotated copy of O_0 lies in one of them.* Then the choice

$$O(X') = R(-\phi)O_0 \tag{31}$$

works for $X' = R(-\phi)X$, with a common angle bound $\delta > 0$. Now fix $\chi \in \mathcal{S}_N$ supported in O_0, and consider the set

$$\{\theta \in \mathbb{R} : D_N[\chi](\theta) = 0\}. \tag{32}$$

This is closed (because $\theta \mapsto D_N[\chi](\theta)$ is continuous) and contains $\theta = 0$. To see that it is open, hence all of the real line, suppose that θ_0 lies in this set and θ satisfies $|\theta - \theta_0| < \delta$. By definition,

$$D_N[\chi](\theta) = D_N[R(\theta_0)^* \chi](\theta - \theta_0) + D_N[\chi](\theta_0). \tag{33}$$

Both terms on the right-hand side are zero. □

The rest of this paper is devoted to the proofs of Lemma 2.1 and Theorem C.

3. Proof sketch

We prove Theorem C by excluding every nonconstant angular Fourier coefficient near narrow configurations. Expanding in Fourier series, write

$$D_N[\chi](\theta) = \sum_{\ell \in \mathbb{Z}} c_\ell^{(N)}[\chi] e^{i\ell\theta}, \qquad c_\ell^{(N)}[\chi] = \frac{1}{2\pi} \int_0^{2\pi} D_N[\chi](\theta) e^{-i\ell\theta} \, \mathrm{d}\theta. \tag{34}$$

It suffices to show that $c_\ell^{(N)} = 0$ for $\ell \neq 0$: the remaining constant is zero because $D_N[\chi](0) = 0$.†

*Apply compactness to the parameter circle: each angle has an interval on which the corresponding rotated neighborhood remains in one fixed open set. A finite subcover gives both a common shrinkage of O_0 and a positive minimum angle bound.

†We suppress N, χ when possible. For real tests, $c_{-\ell}[\chi] = \overline{c_\ell[\chi]}$.

The first step bounds the possible frequencies. Reflection positivity across $t = 0$ constructs the Osterwalder–Schrader Hilbert space and the time and spatial translation generators H, P. Comparing this construction with the one for $t + x = 0$ gives $H \geq |P|$. At a spectral point (ω, p) this says $|p| \leq \omega$, so

$$|e^{-t\omega+ixp}| \leq e^{-\omega(\Re t-|\Im x|)}.$$

Thus translations extend holomorphically to $q(t, x) := \Re t - |\Im x| > 0$. Write $z_j = (t_j, x_j)$ for the complex planar variables. We represent moments by fields inserted between these complex translations. The regularity bound controls the insertions and, for fixed transverse profiles, bounds the kernel by a constant times $\left(1 + (\min_{j<N} q(z_{j+1} - z_j))^{-1}\right)^N$. The transverse variables remain smeared, so the power is independent of d. For a narrow configuration, a suitable quarter turn puts the rotated successive differences in $q > 0$ at each real angle. Imaginary rotation preserves this domain and satisfies $q(R(-i\beta)z) \geq e^{-|\beta|}q(z)$. The kernel bound therefore contributes $e^{N|\beta|}$. For the tensor action at these complex angles, we use the complexification $\mathcal{T}_{\mathbb{C}}$. The weight assumption gives

$$\|\rho(R(\theta + i\beta))v\| \leq e^{2|\beta|}\|v\|, \qquad v \in \mathcal{T}_{\mathbb{C}}. \tag{35}$$

For two-forms the exponent can be replaced by $|\beta|$. The tensor weights thus contribute at most $e^{2N|\beta|}$. Hence, for tests near narrow configurations,

$$|D_N[\chi](\theta + i\beta)| \leq C_\chi e^{3N|\beta|}. \tag{36}$$

Shifting the Fourier integral to height β gives $|c_\ell[\chi]| \leq C_\chi e^{3N|\beta|+\ell\beta}$. Choose the sign of β opposite to ℓ and let $|\beta| \to \infty$: every coefficient with $|\ell| > 3N$ vanishes. The quarter turn is already a symmetry, so only $\ell \in 4\mathbb{Z}$ can occur.

At order N, let M_N be the largest absolute value of a nonzero frequency occurring near some narrow configuration, and set $M_N = 0$ if there is none. Thus $M_0 = 0$ and $0 \leq M_N \leq 3N$. To eliminate the remaining frequencies, we compare moments of orders $a + b$, $2a$ and $2b$ by reflection positivity. A narrow configuration can be rotated and split by the reflection plane into groups of any prescribed sizes a, b. The mixed reflected tuple and the two tuples defining its squared norms all remain narrow. Consequently the same Fourier bounds apply to all three; §7.1 gives the precise support conditions.

This comparison must hold at imaginary rotation angles, where the Fourier frequencies become exponential growth rates. We therefore continue the positive reflection form itself, using Lemma 5.1. For products A, B of a, b fields with tuples χ, η as above, the resulting positive form is

$$\langle A, B\rangle_\beta = S_{a+b}(J\chi, \eta) + D_{a+b}[J\chi, \eta](i\beta), \qquad \beta \in \mathbb{R},$$

where $J\chi$ is the reflected, conjugated tuple. If the coefficient at $-M_{a+b}$ is nonzero, the mixed pairing has a nonzero leading term proportional to $e^{M_{a+b}\beta}$ as $\beta \to +\infty$. Its two squared norms grow at most as $e^{M_{2a}\beta}$ and $e^{M_{2b}\beta}$. Cauchy–Schwarz bounds the square of the mixed pairing by their product; comparison of the exponents gives

$$2M_{a+b} \leq M_{2a} + M_{2b}. \tag{37}$$

This is the convexity inequality used below.

There is a second restriction on the even orders. Suppose $M_{2a} > 0$ and restrict the continued form to products of a fields of the type just described. Multiplying it by $e^{-M_{2a}\beta}$ and letting $\beta \to +\infty$ gives a nonzero positive semidefinite form whose entries are the leading coefficients. Some diagonal coefficient is therefore strictly positive: otherwise Cauchy–Schwarz would make every entry zero. Now rotate the law through 45°. This permutes the four required reflection hyperplanes, so the rotated law satisfies the same hypotheses and its corresponding diagonal coefficient is nonnegative. But that coefficient is multiplied by $e^{-iM_{2a}\pi/4} = (-1)^{M_{2a}/4}$. Its sign can be preserved only if $M_{2a} \in 8\mathbb{Z}$.

We therefore prove the three conditions

$$M_N \le 3N, \qquad 2M_{a+b} \le M_{2a} + M_{2b}, \qquad M_{2a} \in 8\mathbb{Z}.$$

Together with $M_0 = 0$ and nonnegativity, these force every M_N to vanish by Lemma 8.1. This gives Theorem C; the geometric reductions above then give the main theorem.

4. Consequences of reflection positivity (RP)

4.1. **The Osterwalder–Schrader Hilbert space.** The reflected quotient and translation semigroup follow the Osterwalder–Schrader construction [OS73, §4.1, (4.3)–(4.9)]. We use the ambient $L^2(\mu)$ bound in the contraction argument and recall the details without assuming continuous rotation covariance.

Here we complexify the observable space to $L^2(\mathcal{S}', \mu)_{\mathbb{C}}$. Let $L^2_{+,\mathbb{C}}$ denote its closed subspace generated by observables depending on fields in the forward half-space $\{t > 0\}$. In particular, it contains all products

$$\mathrm{eval}_\chi(F) = F(\chi_1) \cdots F(\chi_N) \tag{38}$$

with the tests supported there, and $\mathrm{eval}_\varnothing = 1$. We include the constant and allow complex coefficients in the algebra of such products. These polynomials are dense in $L^2_{+,\mathbb{C}}$.*

Write $\Theta_\circ(t, x, y) = (-t, x, y)$ and set $J_\circ A = \overline{A(\Theta_\circ^* F)}$. Reflection invariance makes $J_\circ$ an antiunitary involution on $L^2(\mu)_{\mathbb{C}}$. Define

$$\langle A, B\rangle_\circ = \mathbb{E}_\mu[(J_\circ A)B]. \tag{39}$$

The form is antilinear in its first argument. Reflection positivity is equivalent to its being positive semidefinite on $L^2_{+,\mathbb{C}}$. Indeed, on field exponentials its matrix is the transpose of the characteristic-functional matrix in Item (iii), and the transpose of a positive Hermitian matrix is positive. The estimate

$$|\langle A, B\rangle_\circ| \le \|A\|_{L^2(\mu)} \|B\|_{L^2(\mu)} \tag{40}$$

extends positivity to the closure. Conversely, applying positivity to finite sums of exponentials gives the original condition.

Quotient by the null space and complete to obtain $\mathcal{H}_{\circ,\mathbb{C}}$. We conflate an observable with its class in this space, and put $\Omega = 1$. The quotient map is a contraction. Unmarked observable norms below are quotient norms; ordinary probability-space norms carry an $L^p(\mu)$ subscript.

Let $T_{\delta t, \delta x} : (t, x, y) \mapsto (t + \delta t, x + \delta x, y)$ denote a translation fixing y. Its action on observables will mean $U(\delta t e_0 + \delta x e_1)$. For $s \ge 0$, reflection and translation give

$$\|U(se_0)A\|^2 = \langle A, U(2se_0)A\rangle \le \|A\| \, \|U(2se_0)A\|. \tag{41}$$

Iterating and using $\|U(re_0)A\| \le \|A\|_{L^2(\mu)}$ proves contractivity and preservation of null vectors. The induced operators are self-adjoint and strongly continuous. Translations parallel to $t = 0$ induce commuting strongly continuous unitary groups. Therefore there are a nonnegative self-adjoint H and a self-adjoint P with joint projection-valued measure Π such that

$$V(\delta t, \delta x) = e^{-\delta t H + i \delta x P} = \int_{\mathbb{R}^2} e^{-\omega \delta t + i p \delta x} \, \mathrm{d}\Pi(\omega, p), \qquad \delta t \ge 0. \tag{42}$$

They implement the translations of classes. Strong continuity follows first on field exponentials from continuity of their tests and the field moment bound, and then by density. The vacuum is fixed by every translation. The same construction applies with any of the assumed reflection hyperplanes as time zero.

*The regularity bound allows the Taylor polynomials of $e^{iF(\chi)}$ to converge in L^2. Field exponentials have dense span: a function orthogonal to them has finite-dimensional conditional measures with zero Fourier transform, hence zero conditional expectations. Evaluations generate the sigma-algebra; see §A.

4.2. **Spectral inequality.** Write $\Theta_\Delta(t,x,y) = (-x,-t,y)$ and J_Δ for its antilinear reflection. Now consider the Hamiltonian H and the x-momentum generator P, constructed in the previous subsection on the Osterwalder–Schrader Hilbert space.

Proposition 4.1. $H \geq |P|$ ■

Proof. Let $A \in L^2_{+,\mathbb{C}}$ denote a real polynomial in compactly supported field variables and $T = U(\delta t e_0 + \delta x e_1)$ a translation with $\delta t > 0$. Consider the quantity

$$\mathbb{E}_\mu[(J_\circ A)TA] = \langle A, TA\rangle_{\mathcal{H}_{\circ,\mathbb{C}}}. \tag{43}$$

Letting $\rho = \langle A, \Pi(-)A\rangle_{\mathcal{H}_{\circ,\mathbb{C}}} : \mathrm{Borel}(\mathbb{R}^2) \to [0,\infty)$ denote the spectral measure associated to A, the right-hand side above is

$$\langle A, TA\rangle_{\mathcal{H}_{\circ,\mathbb{C}}} = \int_{\mathbb{R}^2} e^{-\omega\delta t + ip\delta x}\,\mathrm{d}\rho(\omega,p). \tag{44}$$

Our goal is to show that ρ is supported in the region $\{\omega \geq |p|\}$. Because the support of the spectral measure does not change upon translating A forward in time, it suffices to replace A by a forward translate by a fixed $b > 0$. Since our tests are compactly supported, by choosing b large enough we may arrange that A is supported in $\{t > |x|\}$.

Then A and $J_\Delta J_\circ A$ are both tests in the second Osterwalder–Schrader Hilbert space $\mathcal{H}_{\Delta,\mathbb{C}}$. Thus, if T maps the half space $\{t + x > 0\}$ to itself, then

$$\mathbb{E}_\mu[(J_\circ A)TA] = \mathbb{E}_\mu[J_\Delta(J_\Delta J_\circ A)TA] = \langle J_\Delta J_\circ A, TA\rangle_{\mathcal{H}_{\Delta,\mathbb{C}}} \tag{45}$$

may be interpreted as an inner product in $\mathcal{H}_{\Delta,\mathbb{C}}$. The condition that $T_{\delta t,\delta x}$ map $\{t + x > 0\}$ to itself says that $\delta t + \delta x > 0$. Put $v = t + x$ and $u = t - x$. In $\mathcal{H}_{\Delta,\mathbb{C}}$, the Hamiltonian H_Δ is the generator of translations in v, and P_Δ is the generator of translations in u. Hence

$$T = \int_{\mathbb{R}^2} e^{-\omega(\delta t+\delta x)+ip(\delta t-\delta x)}\,\mathrm{d}\Pi_\Delta(\omega,p) = \int_{\mathbb{R}^2} e^{(ip-\omega)\delta t-(ip+\omega)\delta x}\,\mathrm{d}\Pi_\Delta(\omega,p) \tag{46}$$

where $\Pi_\Delta : \mathrm{Borel}(\mathbb{R}^2) \to \mathcal{B}(\mathcal{H}_{\Delta,\mathbb{C}})$ is the spectral measure for the pair (H_Δ, P_Δ). So, letting $\rho_\Delta = \langle J_\Delta J_\circ A, \Pi_\Delta A\rangle_{\mathcal{H}_{\Delta,\mathbb{C}}}$ (this is a complex measure), we have

$$\langle J_\Delta J_\circ A, TA\rangle_{\mathcal{H}_{\Delta,\mathbb{C}}} = \int_{\mathbb{R}^2} e^{(ip-\omega)\delta t-(ip+\omega)\delta x}\,\mathrm{d}\rho_\Delta(\omega,p). \tag{47}$$

In summary,

$$\int_{\mathbb{R}^2} e^{-\omega\delta t+ip\delta x}\,\mathrm{d}\rho(\omega,p) = \mathbb{E}_\mu[(J_\circ A)TA] = \int_{\mathbb{R}^2} e^{(ip-\omega)\delta t-(ip+\omega)\delta x}\,\mathrm{d}\rho_\Delta(\omega,p), \tag{48}$$

This holds for all $\delta t > 0$ and $\delta x > -\delta t$. For the argument below, it is somewhat simpler to work with

$$\delta v = \delta t + \delta x, \quad \delta u = \delta t - \delta x. \tag{49}$$

Then, the identity above reads

$$\int_{\mathbb{R}^2} e^{\frac{1}{2}(ip-\omega)\delta v-\frac{1}{2}(\omega+ip)\delta u}\,\mathrm{d}\rho(\omega,p) = \int_{\mathbb{R}^2} e^{-\omega\delta v+ip\delta u}\,\mathrm{d}\rho_\Delta(\omega,p). \tag{50}$$

In order to extract information about the measures ρ, ρ_Δ, we integrate both sides of this equality against a function of the form $h(\delta v)\overline{h(\delta u)}$ for $h \in L^1((0,\infty))_\mathbb{C}$. Recall that

$$\mathcal{L}h(z) = \int_0^\infty e^{-zs}h(s)\,\mathrm{d}s \tag{51}$$

denotes the Laplace transform of h. We now compute

$$\int_0^\infty\int_0^\infty h(\delta v)\overline{h(\delta u)}\int_{\mathbb{R}^2} e^{\frac{1}{2}(ip-\omega)\delta v-\frac{1}{2}(\omega+ip)\delta u}\,\mathrm{d}\rho(\omega,p)\,\mathrm{d}\delta u\,\mathrm{d}\delta v = \int_{\mathbb{R}^2}\Big|\mathcal{L}h\Big(\frac{\omega-ip}{2}\Big)\Big|^2\,\mathrm{d}\rho(\omega,p) \tag{52}$$

$$\int_0^\infty\int_0^\infty h(\delta v)\overline{h(\delta u)}\int_{\mathbb{R}^2} e^{-\omega\delta v+ip\delta u}\,\mathrm{d}\rho_\Delta(\omega,p)\,\mathrm{d}\delta u\,\mathrm{d}\delta v = \int_{\mathbb{R}^2}\mathcal{L}h(\omega)\overline{\mathcal{L}h(ip)}\,\mathrm{d}\rho_\Delta(\omega,p). \tag{53}$$

Thus,

$$0 \le \int_{\mathbb{R}^2} \Big|\mathcal{L}h\Big(\frac{\omega - ip}{2}\Big)\Big|^2 \,\mathrm{d}\rho(\omega,p) = \int_{\mathbb{R}^2} \mathcal{L}h(\omega)\overline{\mathcal{L}h(ip)}\,\mathrm{d}\rho_\Delta(\omega,p) \\ \le \|\rho_\Delta\| \sup_{\omega\ge 0}|\mathcal{L}h(\omega)| \sup_{p\in\mathbb{R}}|\mathcal{L}h(ip)|. \tag{54}$$

By Cauchy–Schwarz, $\|\rho_\Delta\| \le \|J_\Delta J_\circ A\|_{\mathcal{H}_{\Delta,\mathbb{C}}}\|A\|_{\mathcal{H}_{\Delta,\mathbb{C}}} < \infty$.

Now the game is to find h such that $\mathcal{L}h$ is not too large on $\mathbb{R}^+ \cup i\mathbb{R}$ but $\mathcal{L}h$ is large at $(\omega - ip)/2$ whenever $|p| > \omega$. Take $h(s) = s^{iF}\,e^{-\varepsilon s}$, where $\varepsilon > 0$ and $F \in \mathbb{R}$ are parameters. This satisfies

$$\mathcal{L}h(z) = \int_0^\infty s^{iF}\,e^{-s(z+\varepsilon)}\,\mathrm{d}s = C\Big(1 + \frac{z}{\varepsilon}\Big)^{-1-iF} \tag{55}$$

for some numerical constant $C = C(\varepsilon, F) \in \mathbb{C}^\times$. Thus,

$$|\mathcal{L}h(z)| = |C| \cdot \Big|1 + \frac{z}{\varepsilon}\Big|^{-1} e^{F\arg(\varepsilon+z)}. \tag{56}$$

Dividing out $|C|^2$, the inequality above reads

$$\int_{\mathbb{R}^2} \Big|1 + \frac{\omega - ip}{2\varepsilon}\Big|^{-2} e^{2F\arg(\varepsilon+z)}\,\mathrm{d}\rho(\omega,p) \le \|\rho_\Delta\| e^{\pi|F|/2} \tag{57}$$

for $z = (\omega - ip)/2$. The right-hand side is $O(e^{\pi|F|/2})$ as $|F| \to \infty$. But if $(\omega, p) \in \operatorname{supp}\rho$ and $|\arg(\varepsilon+z)| > \pi/4$, then for some $\eta > 0$ the left-hand side is $\Omega(e^{(\pi/2+\eta)|F|})$ as $F \to +\infty$ or $F \to -\infty$.* These are only compatible if $|\arg(\varepsilon + z)| \le \pi/4$, which means $\omega + 2\varepsilon \ge |p|$. Taking $\varepsilon \to 0$, we conclude $\omega \ge |p|$.† □

5. Extending reflection positivity

Reflection positivity is preserved under certain sorts of holomorphic deformations of the reflection form. The lemma below is a one-variable form of Glaser's positive-kernel continuation principle [Gla74, Theorem A]; we give the proof using derivative Gram matrices and his Taylor-expansion argument [Gla69, §3, p. 14].

Lemma 5.1. Let $\mathcal{X}$ denote a vector space, let $j : \mathcal{X}_\mathbb{C} \to L^2_{+,\mathbb{C}}$ be a map on its complexification, and let $O = O^* \subset \mathbb{C}$ denote a connected open domain in the complex plane containing the origin. Say we have chosen, for each pair of complex numbers $\xi, \zeta \in O$, a sesquilinear form $\langle -,-\rangle_{\zeta,\xi} : \mathcal{X}^2_\mathbb{C} \to \mathbb{C}$ such that

- for each $A, B \in \mathcal{X}_\mathbb{C}$, the function $(\zeta,\xi) \mapsto \langle A, B\rangle_{\zeta,\xi}$ is jointly holomorphic on O^2;
- for each $X \in \mathcal{X}_\mathbb{C}$ there exists a $\delta > 0$ and smooth curve $[0,\delta) \ni s \mapsto X(s) \in L^2_{+,\mathbb{C}}$ such that $X(0) = j(X)$ and
$$\langle A, B\rangle_{r,s} = \langle A(r), B(s)\rangle_\circ \tag{58}$$
for all $r, s > 0$ small enough.

Then, for each $z \in O$, the form $\langle -,-\rangle_{z,\bar z}$ is positive semidefinite:

$$\langle A, A\rangle_{z,\bar z} \ge 0 \tag{59}$$

for all $A \in \mathcal{X}_\mathbb{C}$. ■

Here $O^* = \{\bar z : z \in O\}$, and the forms are antilinear in their first entry, consistently with (39).

*Choose a neighborhood of the point on which the argument is everywhere greater than $\pi/4+\eta/2$ or everywhere less than $-\pi/4-\eta/2$. Its ρ-measure is positive, as is its measure with the strictly positive weight $|1+z/\varepsilon|^{-2}$. Restricting the integral to this neighborhood gives the stated lower bound for the corresponding sign of F.

†The preliminary forward translation multiplies the spectral measure by $e^{-2b\omega} > 0$, so its support is unchanged. Polynomial classes with compact positive-time supports have dense complex span; hence the spectral projection of $\{\omega < |p|\}$ vanishes on $\mathcal{H}_{\circ,\mathbb{C}}$.

Proof idea. Fix $A \in \mathcal{X}_{\mathbb{C}}$. To warm up, we show that $\langle A, A\rangle_{z,\bar{z}} \geq 0$ for z near the origin. Via analyticity, we may Taylor expand

$$\langle A, A\rangle_{\zeta,\xi} = \sum_{a=0}^{\infty}\sum_{b=0}^{\infty} \frac{\zeta^a \xi^b}{a!b!} \partial_r^a \partial_s^b \langle A, A\rangle_{r,s}\Big|_{r=s=0} \tag{60}$$

for ζ, ξ near 0. The series on the right-hand side is absolutely convergent. Now plug in $\zeta = z$ and $\xi = \bar{z}$:

$$\begin{aligned} \langle A, A\rangle_{z,\bar{z}} &= \sum_{a=0}^{\infty}\sum_{b=0}^{\infty} \frac{z^a \bar{z}^b}{a!b!} \partial_r^a \partial_s^b \langle A, A\rangle_{r,s}\Big|_{r=s=0} \\ &= \lim_{N\to\infty} \sum_{a,b=0}^{N} \frac{z^a \bar{z}^b}{a!b!} \partial_r^a \partial_s^b \langle A, A\rangle_{r,s}\Big|_{r=s=0}. \end{aligned} \tag{61}$$

We can write the quantity under the limit as

$$\sum_{a,b=0}^{N} \frac{z^a \bar{z}^b}{a!b!} \partial_\zeta^a \partial_\xi^b \langle A, A\rangle_{\zeta,\xi}\Big|_{\zeta=\xi=0} = v^\dagger M[N](0) v \tag{62}$$

for $v = (v_0, \ldots, v_N) \in \mathbb{C}^{N+1}$ the vector whose jth component is $v_j = \bar{z}^j/j!$ and $M[N](z)$ the generally complex matrix

$$M[N](z) = \begin{bmatrix} \langle A, A\rangle_{\zeta,\xi} & \cdots & \partial_\xi^N \langle A, A\rangle_{\zeta,\xi} \\ \vdots & \ddots & \vdots \\ \partial_\zeta^N \langle A, A\rangle_{\zeta,\xi} & \cdots & \partial_\zeta^N \partial_\xi^N \langle A, A\rangle_{\zeta,\xi} \end{bmatrix}\Bigg|_{\zeta=z,\xi=\bar{z}}. \tag{63}$$

The (a, b)th entry of $M[N](z)$ is

$$M[N]_{a,b}(z) = \partial_\zeta^a \partial_\xi^b \langle A, A\rangle_{\zeta,\xi}\Big|_{\zeta=z,\xi=\bar{z}}, \qquad 0 \leq a, b \leq N. \tag{64}$$

If we know that $M[N](0)$ is positive semidefinite, then $\langle A, A\rangle_{z,\bar{z}} \geq 0$ follows. To see that $M[N](0)$ is positive semidefinite, write

$$\begin{aligned} M[N]_{a,b}(0) &= \partial_r^a \partial_s^b \langle A, A\rangle_{r,s}\Big|_{r=s=0} \\ &= \partial_r^a \partial_s^b \langle A(r), A(s)\rangle_\circ\Big|_{r=s=0} \\ &= \langle \partial_r^a A(r), \partial_s^b A(s)\rangle_\circ\Big|_{r=s=0} = \langle A^{(a)}(0), A^{(b)}(0)\rangle_\circ. \end{aligned} \tag{65}$$

Thus, for every $v \in \mathbb{C}^{N+1}$,

$$v^\dagger M[N](0) v = \Big\langle \sum_{n=0}^{N} v_n A^{(n)}(0), \sum_{n=0}^{N} v_n A^{(n)}(0) \Big\rangle_\circ \geq 0, \tag{66}$$

since $\langle -, -\rangle_\circ$ is positive semidefinite, as desired.* □

Proof. Fix A, and consider the subset $\Omega \subset O$ consisting of $z \in O$ such that, for each $N \geq 0$, the complex matrix $M[N](z)$ defined above is positive semidefinite. Our goal is to show that $\Omega = O$. Indeed, if $M[N](z)$ is positive semidefinite, then

$$\langle A, A\rangle_{z,\bar{z}} = \mathbf{e}^\dagger M[N](z) \mathbf{e} \geq 0 \tag{67}$$

for $\mathbf{e} = (1, 0, \ldots, 0)$.

*One-sided derivatives at zero suffice. Smoothness of the curves and continuity of the reflection pairing justify differentiation; the derivatives remain in the closed space $L^2_{+,\mathbb{C}}$.

Because the matrix $M[N](z)$ depends continuously on z, the set Ω is closed in O. The sketch above contains the full proof that $0 \in \Omega$, so Ω is nonempty. Next, we show that Ω is open, hence all of O since O is connected. This is the general complex analytic fact in Theorem D, applied to the holomorphic function $F(\zeta, \xi) = \langle A, A \rangle_{\zeta,\xi}$. □

The following is the one-variable derivative-positivity statement in Glaser's argument [Gla69, §3, p. 14].*

Theorem D. Let $O \subset \mathbb{C}$ be open and invariant under complex conjugation, and let $F : O^2 \to \mathbb{C}$ be holomorphic. For $N \geq 0$ define

$$M_F[N](z)_{a,b} = \partial_\zeta^a \partial_\xi^b F(\zeta, \xi)\Big|_{\zeta=z,\xi=\bar{z}}, \qquad 0 \leq a, b \leq N. \tag{68}$$

Then the set

$$\Omega_F = \{z \in O : M_F[N](z) \text{ is positive semidefinite for every } N \geq 0\} \tag{69}$$

is open in O. ■□

Proof. Suppose $z_0 \in \Omega_F$. Taylor expand

$$\partial_\zeta^a \partial_\xi^b F(\zeta, \xi) = \sum_{\alpha,\beta \geq 0} \frac{(\zeta - z_0)^\alpha (\xi - \bar{z}_0)^\beta}{\alpha! \beta!} \partial_\zeta^{a+\alpha} \partial_\xi^{b+\beta} F(z_0, \bar{z}_0) \tag{70}$$

on a sufficiently small polydisc about $(z_0, \bar{z}_0)$. This series is absolutely convergent. If z is sufficiently close to z_0, it follows that

$$M_F[N](z) = \lim_{L \to \infty} Z_L M_F[N + L](z_0) Z_L^\dagger, \tag{71}$$

where Z_L is the $(N+1)$-by-$(N+L+1)$ matrix with entries

$$(Z_L)_{a,c} = \begin{cases} (z - z_0)^{c-a}/(c-a)!, & 0 \leq c - a \leq L, \\ 0, & \text{otherwise.} \end{cases} \tag{72}$$

Indeed, the (a, b) entry of the matrix under the limit is

$$\sum_{\alpha,\beta=0}^{L} \frac{(z - z_0)^\alpha (\bar{z} - \bar{z}_0)^\beta}{\alpha! \beta!} \partial_\zeta^{a+\alpha} \partial_\xi^{b+\beta} F(z_0, \bar{z}_0), \tag{73}$$

which converges to $M_F[N](z)_{a,b}$. Each $Z_L M_F[N + L](z_0) Z_L^\dagger$ is the congruence of a positive semidefinite matrix, hence positive semidefinite itself. Therefore, for every $v \in \mathbb{C}^{N+1}$,

$$v^\dagger M_F[N](z) v = \lim_{L \to \infty} (Z_L^\dagger v)^\dagger M_F[N + L](z_0)(Z_L^\dagger v) \geq 0. \tag{74}$$

The same polydisc works for every N, since all derivatives of F are holomorphic there. Thus every $M_F[N](z)$ is positive semidefinite for z in one common neighborhood of z_0. Consequently $z \in \Omega_F$, and Ω_F is open. □

*The same Taylor re-expansion is given in [BEM98, Appendix C, (145)–(146)].

6. The analytic continuation

The bound $|P| \le H$ lets positive-time damping compensate for the growth of imaginary spatial translations. Their spectral multipliers have modulus at most $e^{-\omega(\Re t - |\Im x|)}$, so the quantity below determines the complex translation domain.

For $z = (t, x) \in \mathbb{C}^2$ let

$$q(z) = \Re t - |\Im x|, \qquad \mathcal{W} = \{z : q(z) > 0\}. \tag{75}$$

By Proposition 4.1, the translation operators extend to

$$V(z) = \int e^{-t\omega + ixp} \, \mathrm{d}\Pi(\omega, p). \tag{76}$$

They are contractions on $q \ge 0$, strongly continuous there and operator-norm holomorphic on $\mathcal{W}$. Indeed, on $q \ge \delta > 0$ a spectral derivative of total order k is bounded by $\omega^k e^{-\delta\omega} \le k!\delta^{-k}$, since $|p| \le \omega$. The semigroup law still holds, and $V(z)\Omega = V(z)^*\Omega = \Omega$.

6.1. **Analyticity across a separating hyperplane.** We prove Lemma 2.1. Translate its separating hyperplane to time zero, using its coordinate normal as the time direction. Choose $\delta > 0$ and the supports sufficiently close to the given configuration that the negative group lies in $t < -\delta$ and the positive group in $t > \delta$. Let A_- and A_+ denote the products of fields on the negative and positive sides. Put

$$\psi_- = U(-\delta\mathbf{n})JA_-, \qquad \psi_+ = U(-\delta\mathbf{n})A_+, \tag{77}$$

where J is reflection in this hyperplane. Both define classes in the Osterwalder–Schrader Hilbert space, and reflection and translation give

$$\mathbb{E}_\mu[A_- U(s\mathbf{n} + \mathbf{v})A_+] = \langle \psi_-, V(2\delta + s, \mathbf{v})\psi_+ \rangle. \tag{78}$$

The ordinary pullback in the lemma is $T^*_{-s\mathbf{n}-\mathbf{v}}$, exactly the translation $U(s\mathbf{n} + \mathbf{v})$ here.

For a coordinate normal, repeat the cone argument with each other coordinate as the transverse direction. The joint spectral variables then satisfy $|p_i| \le \omega$ for every tangential coordinate. Consequently the right side of (78) is holomorphic whenever

$$2\delta + \Re s > \sum_i |\Im v_i|. \tag{79}$$

This proves the claimed real analyticity for $s > -2\delta$ and all real $\mathbf{v}$. Under the four planar hypotheses the same argument uses only the other planar momentum, proving the last assertion of the lemma. If one of the groups is empty the moment is constant by translation invariance. This completes the proof of Lemma 2.1.

6.2. **A field in a time slab.** We need to insert a field between translations with a bound in the Osterwalder–Schrader quotient norm. The ordinary $L^2(\mu)$ moment bound does not by itself give this operator estimate. We first bound an exponential insertion by repeated reflection, then differentiate in its source. Below K depends only on the constant C in (4) and may increase between estimates.

Repeated reflection doubles the time slab, the L^1 norm and the squared L^2 norm of the source at each step. Cauchy–Schwarz takes the corresponding 2^{-r} power, leaving the exponential bound independent of the number of reflections.* We now extend F complex-linearly to $(L^1 \cap L^2)(\mathbb{R}^d; \mathcal{T})_{\mathbb{C}}$ and its moments complex-multilinearly. Let h belong to this space, supported in $0 \le t \le l$, and let A be a bounded positive-time observable. We claim

$$\|e^{F(h)} U(le_0)A\| \le e^{C(\|\Re h\|_1 + \|\Re h\|_2^2)} \|A\|. \tag{80}$$

*We adapt the dyadic Cauchy–Schwarz iteration in Seiler–Simon [SS76, Theorem 4.3, (23)–(24), p. 490]. Their proof of Theorem 7.1, (31)–(33), pp. 502–503, duplicates reflected sources; the proof of Theorem 8.2, pp. 509–510, combines reflections with Cauchy estimates for moments. Those results concern specific models; here the terminal bound is (4).

To prove this, reflection followed by translation gives

$$\|e^{F(h)}U(le_0)A\|^2 = \langle A, e^{F(\tau_{le_0}(Jh+h))}U(2le_0)A\rangle, \tag{81}$$

where on tests $Jh = \Theta_\circ^* \bar{h}$. Define

$$h_0 = h, \quad h_{r+1} = \tau_{2^r le_0}(Jh_r + h_r), \quad a_r = \|e^{F(h_r)}U(2^r le_0)A\|. \tag{82}$$

The source h_r is supported in $0 \le t \le 2^r l$. Since the adjacent slab interiors are disjoint,

$$\|\Re h_r\|_1 = 2^r\|\Re h\|_1, \qquad \|\Re h_r\|_2^2 = 2^r\|\Re h\|_2^2.$$

Cauchy–Schwarz gives

$$a_r^2 \le \|A\| a_{r+1}, \qquad a_0 \le \|A\|^{1-2^{-r}} a_r^{2^{-r}}. \tag{83}$$

On the other hand, the defining reflection form and (4) give

$$a_r^2 \le \|A\|_{L^\infty(\mu)}^2 \exp(C2^{r+1}(\|\Re h\|_1 + \|\Re h\|_2^2)). \tag{84}$$

Taking the square root and then the 2^{-r} power leaves the exponent $C(\|\Re h\|_1 + \|\Re h\|_2^2)$, while $\|A\|_{L^\infty(\mu)}^{2^{-r}}$ tends to one. The surviving dependence on A is $\|A\|$, giving (80) as $r \to \infty$. If $\|A\| = 0$, the first Cauchy–Schwarz inequality already gives $a_0 = 0$.

For χ supported in $|t| \le \delta$, define

$$B_\delta(\chi)A = F(\tau_{\delta e_0}\chi)U(2\delta e_0)A. \tag{85}$$

Apply Cauchy's estimate to the entire Hilbert-valued map $z \mapsto e^{zF(\tau_{\delta e_0}\chi)}U(2\delta e_0)A$ on $|z| = (\|\chi\|_1 + \|\chi\|_2)^{-1}$ for $\chi \ne 0$. Its power series converges first in ordinary $L^2(\mu)_\mathbb{C}$, by (4). Equation (80) therefore gives

$$\|B_\delta(\chi)\| \le K(\|\chi\|_1 + \|\chi\|_2). \tag{86}$$

The estimate shows that the insertion vanishes on null vectors and therefore defines a bounded operator on $\mathcal{H}_{\circ,\mathbb{C}}$.* Reflection and translation give

$$B_\delta(\chi)^* = B_\delta(J\chi), \tag{87}$$

$$V(r+t,x)B_\delta(\chi)V(r-t,-x) = B_{\delta+r}(\tau_{(t,x,0)}\chi), \qquad r \ge |t|. \tag{88}$$

For example, both sides of (88) applied to A equal $F(\tau_{(\delta+r+t,x,0)}\chi)U(2(\delta+r)e_0)A$. If χ is supported in $|t-c| \le \delta_0$ and $|c| + \delta_0 \le \delta$, the same identity gives

$$B_\delta(\chi) = e^{-(\delta+c-\delta_0)H}B_{\delta_0}(\tau_{-ce_0}\chi)e^{-(\delta-c-\delta_0)H}. \tag{89}$$

6.3. **Localizing an insertion in the plane.** Fix a transverse profile $f \in (L^1 \cap L^2)(\mathbb{R}^{d-2}; \mathcal{T}_\mathbb{C})$. For $d = 2$ this means $f \in \mathcal{T}_\mathbb{C}$, with both norms equal to $|f|$. For planar $z = (t,x)$ put

$$j_\varepsilon(z) = (\pi\varepsilon^2)^{-1}1_{\{|z|<\varepsilon\}}, \qquad A_{f,\varepsilon}(\xi,\eta) = V(\xi - \varepsilon e_0)B_\varepsilon(j_\varepsilon f)V(\eta - \varepsilon e_0). \tag{90}$$

This is defined for $q(\xi), q(\eta) > \varepsilon$. The slab estimate alone gives the bound $K(\|f\|_1 + \varepsilon^{-1}\|f\|_2)$, which diverges as $\varepsilon \downarrow 0$. We use holomorphy to bound the norm in terms of $q_0 = \min(q(\xi), q(\eta))$ instead. We claim

$$\|A_{f,\varepsilon}(\xi,\eta)\| \le K\left(\|f\|_1 + \frac{\|f\|_2}{q_0 - \varepsilon}\right). \tag{91}$$

We average translates of the insertion against a planar L^2 test. Convolution by j_ε contracts both L^1 and L^2; the holomorphic mean-value inequality then recovers the value at the center. Put $r = (q_0 - \varepsilon)/2$ and, for unit vectors u, v, consider $h_\varepsilon(\zeta) = \langle u, A_{f,\varepsilon}(\xi + \zeta, \eta - \zeta)v\rangle$. It is holomorphic for $|\Re\zeta_t| < r$, $|\Im\zeta_x| < r$. Write $\zeta = z + ib$, with $z, b \in \mathbb{R}^2$. For $|b_x| < r$ the outer factors

$$V(\xi - (r+\varepsilon)e_0 + ib), \qquad V(\eta - (r+\varepsilon)e_0 - ib) \tag{92}$$

*Truncating A in $L^4(\mu)_\mathbb{C}$ extends (85) to the products of fields used below. The field L^4 bound and Hölder's inequality justify the passage to the quotient.

are contractions, independent of z. Between them, (88) identifies the remaining operator as $B_{r+\varepsilon}((\tau_z j_\varepsilon)f)$ for $z \in (-r,r)^2$. Thus, for $\alpha \in L^2((-r,r)^2)_{\mathbb{C}}$,

$$\begin{aligned}\left|\int \alpha(z) h_\varepsilon(z+ib)\,\mathrm{d}z\right| &\le K(\|f\|_1\|\alpha * j_\varepsilon\|_1 + \|f\|_2\|\alpha * j_\varepsilon\|_2)\\ &\le K(2r\|f\|_1 + \|f\|_2)\|\alpha\|_2,\end{aligned} \tag{93}$$

where α is extended by zero and $\|\alpha\|_1 \le 2r\|\alpha\|_2$. By duality, the $L^2((-r,r)^2)_{\mathbb{C}}$ norm of $z \mapsto h_\varepsilon(z+ib)$ is at most $K(2r\|f\|_1 + \|f\|_2)$. The mean-value inequality on a complex polydisc of radius $r' < r$, followed by integration over the imaginary square, gives

$$|h_\varepsilon(0)|^2 \le (\pi r'^2)^{-2}(2r)^2 K^2 (2r\|f\|_1 + \|f\|_2)^2. \tag{94}$$

Let $r' \uparrow r$ and take the supremum over u, v to prove (91).

We next identify the limit as $\varepsilon \downarrow 0$. Fix $\delta = q_0/2$, take $0 < \varepsilon < \delta/2$, and let α be supported in $(-\delta/2, \delta/2)^2$. Translation covariance gives

$$\int \alpha(z) A_{f,\varepsilon}(\xi + z, \eta - z)\,\mathrm{d}z = V(\xi - \delta e_0) B_\delta((\alpha * j_\varepsilon)f) V(\eta - \delta e_0). \tag{95}$$

The right side converges in operator norm because $\alpha * j_\varepsilon \to \alpha$ in $L^1 \cap L^2$. For fixed vectors u, v, apply Montel's theorem to $\langle u, A_{f,\varepsilon}(\xi+\zeta, \eta-\zeta)v\rangle$. These scalar functions are locally uniformly bounded by (91). The norm limit in (95) fixes the integral of every subsequential limit against every test on the real square. Thus any two limits agree there, and holomorphic uniqueness makes them agree throughout the polydisc. Every sequence $\varepsilon \downarrow 0$ therefore has the same scalar limit at $\zeta = 0$. The limits are bounded sesquilinear forms in u, v, hence define a bounded operator $A_f(\xi, \eta)$ satisfying

$$\|A_f(\xi,\eta)\| \le K\left(\|f\|_1 + \frac{\|f\|_2}{\min(q(\xi), q(\eta))}\right). \tag{96}$$

It is complex-linear in f. This insertion is localized only in the plane; its transverse variables remain smeared. As $q_0 \downarrow 0$, the L^1 term stays bounded and the L^2 term grows like q_0^{-1}, independently of the number of transverse variables.

The translation and adjoint identities pass to the weak limit:

$$V(\kappa) A_f(\xi,\eta) V(\kappa') = A_f(\kappa + \xi, \eta + \kappa'), \tag{97}$$

$$A_f(\xi,\eta)^* = A_{Jf}(-\Theta_\circ \bar\eta, -\Theta_\circ \bar\xi). \tag{98}$$

Here $q(\kappa), q(\kappa') \ge 0$, and $Jf = \rho(\Theta_\circ)\bar f$ acts only on indices. The shift identity also proves norm holomorphy: near (ξ_0, η_0), choose small $a > 0$ and factor the operator as

$$V(\xi - \xi_0 + ae_0) A_f(\xi_0 - ae_0, \eta_0 - ae_0) V(\eta - \eta_0 + ae_0). \tag{99}$$

For compactly supported $\alpha \in L^2(\mathbb{R}^2)$ with $|t| < \delta$ on its support, we recover the smeared insertion as

$$B_\delta(\alpha f) = \int \alpha(z) A_f(\delta e_0 + z, \delta e_0 - z)\,\mathrm{d}z. \tag{100}$$

The integral converges in operator norm by (96). To verify the identity, take ε smaller than half the distance of the support to $|t| = \delta$. Equation (88) gives the regularized identity

$$\int \alpha(z) A_{f,\varepsilon}(\delta e_0 + z, \delta e_0 - z)\,\mathrm{d}z = B_\delta((\alpha * j_\varepsilon)f). \tag{101}$$

The right side converges in norm to $B_\delta(\alpha f)$. On the left, the matrix elements converge under a uniform bound on the compact support, giving (100). The product below depends on its two middle arguments only through their sum: if $a, b, a', b' \in \mathcal{W}$ and $a + b = a' + b'$, then

$$A_f(\xi, a) A_g(b, \eta) = A_f(\xi, a') A_g(b', \eta). \tag{102}$$

Subtract a sufficiently small δe_0 from a, b, a', b' and apply (97). Both sides equal $A_f(\xi, \delta e_0)V(a + b - 2\delta e_0)A_g(\delta e_0, \eta)$.

6.4. **The ordered moment kernels.** Composing the bounded insertions with successive translation arguments gives a holomorphic kernel for the original ordered moments. To write it, first take tests of the form

$$\chi_j(t, x, y) = \alpha_j(t, x) f_j(y), \qquad f_j \in (L^1 \cap L^2)(\mathbb{R}^{d-2}; \mathcal{T}_{\mathbb{C}}). \tag{103}$$

Finite sums of these products suffice by density in $L^1 \cap L^2$.* For $N \geq 2$ and $Z = (z_1, \dots, z_N)$ with $\Delta_j = z_{j+1} - z_j \in \mathcal{W}$, define

$$S_N(Z; f) = \langle \Omega, A_{f_1}(\kappa, \Delta_1/2) \Big[\prod_{j=2}^{N-1} A_{f_j}(\Delta_{j-1}/2, \Delta_j/2) \Big] A_{f_N}(\Delta_{N-1}/2, \kappa)\Omega \rangle. \tag{104}$$

The product is ordered by increasing j and is the identity if $N = 2$. The choice of $\kappa \in \mathcal{W}$ does not matter: the shift identity and $V\Omega = V^*\Omega = \Omega$ let any two choices be enlarged to a common positive-time value. Put $S_0 = 1$ and $S_1(z; f) = \langle m, \int f(y)\, \mathrm{d}y \rangle$, with m as in §1.1; in $d = 2$ this is $\langle m, f \rangle$.

These kernels are holomorphic in the points, invariant under common translations, and complex multilinear in the profiles. Taking κ with a sufficiently large time component in (96) proves

$$|S_N(Z; f)| \leq K^N \prod_{j=1}^{N} \left(\|f_j\|_1 + q_*^{-1} \|f_j\|_2 \right), \qquad q_* := \min_{j<N} q(z_{j+1} - z_j). \tag{105}$$

For compact planar supports strictly ordered in time, they represent the original moments:

$$S_N(\alpha_1 f_1, \dots, \alpha_N f_N) = \int \prod_j \alpha_j(z_j) S_N(Z; f)\, \mathrm{d}Z. \tag{106}$$

To check the identity, enclose the supports in disjoint slabs $(c_j - \delta_j, c_j + \delta_j)$, ordered in j, and translate all of them into positive time. Equations (85) and (89) express the class of the product as

$$e^{-(c_1 - \delta_1)H} B_{\delta_1}(\tau_{-c_1 e_0} \chi_1) \Big[\prod_{j=2}^{N} e^{-(c_j - \delta_j - c_{j-1} - \delta_{j-1})H} B_{\delta_j}(\tau_{-c_j e_0} \chi_j) \Big] \Omega. \tag{107}$$

Insert (100). The semigroup law combines adjacent translation factors with total argument $z_{j+1} - z_j$; (102) assigns half of this argument to each neighboring insertion. Pairing with Ω gives (106). The successive time differences have a positive lower bound on the compact supports, so the integrands are uniformly bounded there. No product of fields at coincident points is used.

6.5. **Joining the ordered domains.** Permuting the entries gives every time ordering. Quarter turns give the corresponding orderings with either t or x as time. On a quarter-turned domain, rotate the points to the original domain and apply the inverse pullback to their profiles. The kernels agree on overlaps: the intersection is convex, and with every point it contains its real part. A nonempty intersection therefore contains a real open set, where both kernels give the original moments by (106). Testing against products of smooth functions and then applying the holomorphic identity theorem gives equality on the intersection.

We continue to denote the joined kernel by $S_N(Z; f)$. Its real domain contains the configurations for which either all times or all x coordinates are distinct. It is symmetric under simultaneous permutations of points and profiles. With $\varrho(\zeta) = \rho(R(\zeta))$ acting on the tensor indices, quarter-turn symmetry reads

$$S_N(R(-\pi/2)Z; \varrho(\pi/2) f) = S_N(Z; f). \tag{108}$$

*The approximation can be made with prescribed planar supports; the moment bound and multilinearity give the passage to general tests.

This is the discrete symmetry already deduced from reflection positivity, not the continuous covariance being proved.

7. The growth estimate near narrow configurations

To bound the angular Fourier frequencies near narrow configurations, we must continue $D_N[\chi]$ to every complex angle and control its growth as the imaginary part tends to either infinity. Small real angles suffice for the later geometric reductions, but the Fourier contour shift requires this full continuation.

We retain the hypotheses of Theorem B. The ordered kernels and the insertion estimates have already been constructed in §6; in particular, (105) gives

$$|S_N(Z;f)| \le K^N \prod_{j=1}^{N} (\|f_j\|_1 + q_*^{-1}\|f_j\|_2), \qquad q_* := \min_{j<N} q(z_{j+1}-z_j).$$

Fix a narrow planar configuration. Choose an ordering, a unit vector n, and $\alpha < \pi/4$ so that all successive displacements lie within angle α of n. Choose small product neighborhoods with compact closures so that the same ordering, direction n, and angle α work throughout them. Consider real product tests

$$\chi_j(t,x,y) = \alpha_j(t,x) f_j(y)$$

with planar supports in these neighborhoods. Narrowness is an open condition.

Proposition 7.1. If the planar tensor rotation weights have absolute value at most two, then for each tuple of tests just described, $D_N[\chi]$ has an entire, 2π-periodic continuation satisfying

$$|D_N[\chi](\theta + i\beta)| \le C_\chi e^{3N|\beta|} \qquad (\theta, \beta \in \mathbb{R}). \tag{109}$$

Consequently $c_\ell[\chi] = 0$ for $|\ell| > 3N$. ∎

Proof. First cover every real angle. Write $n = R(\phi)(1,0)$. For any real θ, choose an integer k with $|\theta - \phi - k\pi/2| \le \pi/4$. Each vector $R(-\theta + k\pi/2)(z_{j+1}-z_j)$ then makes angle less than $\alpha + \pi/4 < \pi/2$ with the positive t-axis. Thus

$$q(R(-\theta + k\pi/2)(z_{j+1}-z_j)) \ge \cos(\alpha + \pi/4)|z_{j+1}-z_j|. \tag{110}$$

The integer k may depend on θ, but this bound does not. On the compact product support its right side has a common positive lower bound, denoted by δ.

Extend each formula to its full vertical strip. The identity

$$q(z) = \min\{\Re(z_t + iz_x), \Re(z_t - iz_x)\}$$

and the rotation convention give

$$(R(-i\beta)z)_t + i(R(-i\beta)z)_x = e^{-\beta}(z_t + iz_x), \qquad (R(-i\beta)z)_t - i(R(-i\beta)z)_x = e^{\beta}(z_t - iz_x).$$

Consequently

$$q(R(-i\beta)z) \ge e^{-|\beta|} q(z) \qquad (q(z) > 0). \tag{111}$$

In particular, imaginary rotations preserve the ordered domain.

For each integer k, the strict real-angle condition is

$$|\Re\zeta - \phi - k\pi/2| < \pi/2 - \alpha. \tag{112}$$

On this vertical strip, define the continuation of $D_N[\chi]$ by

$$\int \prod_j \alpha_j(z_j) S_N\big(R(-\zeta + k\pi/2)Z; \varrho(\zeta - k\pi/2) f\big)\, dZ - S_N(\chi). \tag{113}$$

Here ϱ acts on the profiles' tensor indices only, as in (108). The scalar tests are always evaluated at real points. Each vector $R(-\zeta + k\pi/2)(z_{j+1}-z_j)$ lies in $\mathcal{W}$, by (111). On each compact subset of

the strip, the quantities $q(R(-\zeta + k\pi/2)(z_{j+1} - z_j))$ have a common positive lower bound on the product support. The kernel bounds therefore justify the integral and its holomorphy.

For real ζ, (113) equals $D_N[\chi](\zeta)$: apply the known quarter-turn covariance to the moment, and then use the ordered smearing identity and a change of variables. This uses no covariance under an arbitrary real rotation. The strips (112) cover $\mathbb{C}$ and overlap, since their width $\pi - 2\alpha$ exceeds their center spacing $\pi/2$. Any overlap contains a real interval, where both formulas equal $D_N[\chi]$. The one-variable identity theorem makes the formulas agree on the entire overlap and define a single entire function. Replacing ζ by $\zeta + 2\pi$ and k by $k + 4$ leaves (113) unchanged, proving periodicity.

Estimate the continuation. For $\zeta = \theta + i\beta$, choose k as in (110). Combining (110) and (111) gives

$$q(R(-\theta - i\beta + k\pi/2)(z_{j+1} - z_j)) \geq \delta e^{-|\beta|}.$$

Thus $q_*^{-1} \leq \delta^{-1} e^{|\beta|}$ for the rotated tuple in (105).

This is where representation theory enters. In an invariant norm, the complexified planar rotation splits into weight spaces with factors $e^{im\theta}$. If $|m| \leq 2$, then

$$\|\varrho(\theta + i\beta - k\pi/2) f_j\|_p \leq e^{2|\beta|} \|f_j\|_p, \qquad p = 1, 2. \tag{114}$$

Real rotations preserve both norms. Each factor in (105) is therefore bounded by $e^{3|\beta|}(\|f_j\|_1 + \delta^{-1}\|f_j\|_2)$, uniformly in k. Consequently

$$\left|S_N(R(-\theta - i\beta + k\pi/2)Z; \varrho(\theta + i\beta - k\pi/2) f)\right| \leq K^N \prod_j (\|f_j\|_1 + \delta^{-1}\|f_j\|_2)\, e^{3N|\beta|}. \tag{115}$$

Integrating against the fixed scalar tests in (113) proves (109) with C_χ independent of θ, β.* For a finite sum of such test products of the same order, sum the corresponding constants.

Finally, shift the contour in (34). The continuation is entire and 2π-periodic, so Cauchy's theorem on the rectangle between heights 0 and β gives

$$\begin{aligned} c_\ell[\chi] &= \frac{1}{2\pi} \int_0^{2\pi} D_N[\chi](\theta + i\beta) e^{-i\ell\theta} e^{\ell\beta}\, d\theta, \\ |c_\ell[\chi]| &\leq C_\chi e^{3N|\beta| + \ell\beta}. \end{aligned} \tag{116}$$

Periodicity cancels the vertical sides. Send $\beta \to +\infty$ when $\ell < -3N$ and $\beta \to -\infty$ when $\ell > 3N$. This proves the cutoff. For $N = 0, 1$ the discrepancy is zero by $S_0 = 1$ and the covariance of S_1 established in §1.1. □

Quarter-turn symmetry makes $\theta \mapsto D_N[\chi](\theta)$ $\pi/2$-periodic; holomorphic uniqueness gives the same periodicity for its entire continuation. Hence only $\ell \in 4\mathbb{Z}$ can occur. Since $D_N[\chi](0) = 0$, the cutoff gives, on these narrow tests,

$$D_N[\chi](\zeta) = \sum_{\substack{\ell \in 4\mathbb{Z} \\ 0 < |\ell| \leq 3N}} c_\ell[\chi](e^{i\ell\zeta} - 1). \tag{117}$$

These are the original Fourier coefficients from (34). The moment bound makes each coefficient a bounded multilinear form on $L^1 \cap L^2$ tests, so density of finite sums of product tests extends (117) to all tests with the same small compact planar supports. The later positivity comparisons eliminate the remaining frequencies on narrow configurations. The reductions in §2, using Lemma 2.1, then give $D_N[\chi](\theta) = 0$ outside Δ.

*For example, one may take $C_\chi = K^N \prod_j [\|\alpha_j\|_1(\|f_j\|_1 + \delta^{-1}\|f_j\|_2)] + |S_N(\chi)|$.

7.1. **Testing on narrow configurations.** Cauchy–Schwarz compares the reflected tuples $(J\chi,\eta)$, $(J\chi,\chi)$ and $(J\eta,\eta)$. All three must be narrow for the frequency bounds just proved to apply. The support conditions below ensure this simultaneously. Write

$$\mathcal{C}_+ = \{(t,x)\in\mathbb{R}^2 : t > |x|\}. \tag{118}$$

We use real product tests $\chi_j = \alpha_j f_j$, with $\alpha_j \in C_c^\infty(\mathbb{R}^2;\mathbb{R})$ and real transverse $L^1\cap L^2$ profiles. Call an ordered tuple *admissible* if its compact planar supports $K_j = \operatorname{supp}\alpha_j$ satisfy

$$K_j \subset \mathcal{C}_+, \qquad K_{j+1} - K_j \subset \mathcal{C}_+. \tag{119}$$

The empty tuple is included. All inequalities are strict and hold for every choice of points in the supports.

For two admissible tuples χ,η with planar supports K_j, L_k, the reflected concatenation $(J\chi,\eta)$ is narrow. Reverse the first block to order it from negative to positive time. Its successive displacement vectors are those of the original tuple acted on by $-\Theta_\circ$, which preserves $\mathcal{C}_+$. When both tuples are nonempty, the displacement from the last reflected point to the first unreflected point satisfies

$$z_1' - \Theta_\circ z_1 = z_1' + (-\Theta_\circ z_1) \in \mathcal{C}_+, \qquad z_1\in K_1, \quad z_1'\in L_1, \tag{120}$$

since both summands lie in $\mathcal{C}_+$. Thus every successive displacement vector lies in $\mathcal{C}_+$, including for squared norms and mixed pairings of different admissible tuples. A finite subdivision of the compact supports therefore makes (117) available for all these pairings.

Lemma 7.2. If $\ell\neq 0$ and $c_\ell^{(N)}$ is nonzero near a narrow configuration, then for every $a+b=N$ there are admissible tuples χ,η of lengths a,b such that $c_\ell^{(N)}[J\chi,\eta]\neq 0$. ■

Proof. The identity

$$D_N[R(\phi)^*\chi](\theta) = D_N[\chi](\theta+\phi) - D_N[\chi](\phi) \tag{121}$$

gives, directly from the Fourier integral,

$$c_\ell[R(\phi)^*\chi] = e^{i\ell\phi}c_\ell[\chi], \qquad \ell\neq 0. \tag{122}$$

Common translations and permutations also preserve the coefficients. Localize a nonzero coefficient at a narrow configuration. Apply a common rotation and order the planar points $z_1,\dots,z_N$ so that $z_{j+1}-z_j\in\mathcal{C}_+$. These operations preserve nonvanishing. If both groups are nonempty, choose the origin at the midpoint of z_a, z_{a+1}. For $j\le a$ the vector from z_j to the midpoint is one half of $z_{a+1}-z_a$ plus a sum of successive displacement vectors in $\mathcal{C}_+$; for $j>a$ the vector from the midpoint to z_j has the same property. Consequently the first group lies in $-\mathcal{C}_+$ and the second in $\mathcal{C}_+$. Reflect and reverse the first group. Both tuples now satisfy (119) on sufficiently small product neighborhoods, where real product tests with nonzero coefficient can be chosen.* If one group is empty, put the origin sufficiently far forward or backward in time instead. This proves the claim. □

7.2. **The continued reflection pairing.** The Fourier growth comparison needs a positive form at imaginary rotation angles. We obtain a jointly holomorphic form by rotating the two groups independently; for small real parameters it is their ordinary reflection pairing. Lemma 5.1 then gives positivity at conjugate parameters, where the two groups undergo a common imaginary rotation. Let $A=\prod_{j=1}^a F(\chi_j)$ and $B=\prod_{k=1}^b F(\eta_k)$ be monomials with admissible tuples. In particular, their supports are strictly ordered in $t>0$; the two positive tuples may overlap each other. For real r,s small enough, put

$$A(r) = \prod_j F(R(r)^*\chi_j), \qquad B(s) = \prod_k F(R(s)^*\eta_k). \tag{123}$$

*By the Schwartz kernel theorem, a distribution is nonzero on some product of tests in every product neighborhood of a point in its support. Approximation by finite sums of $\alpha(t,x)f(y)$ in $L^1\cap L^2$, followed by multilinearity and the moment bound, gives the required real product tests.

These are smooth $L^2_{+,\mathbb{C}}$ curves by the moment estimates. Let $\mathcal{X}$ be the real formal span of such monomials. On its complexification $\mathcal{X}_{\mathbb{C}}$, extend the forms sesquilinearly, conjugating the first coefficient. It suffices to work with any fixed finite family.*

Write $\chi_j = \alpha_j f_j$ and $\eta_k = \gamma_k g_k$. Define $\langle A, B\rangle_{\zeta,\xi}$ by integrating the ordered kernel with its first group of points and profiles

$$R(\zeta)\Theta_\circ Z, \quad \varrho(-\zeta)Jf, \qquad \text{and its second group} \qquad R(-\xi)Z', \quad \varrho(\xi)g, \tag{124}$$

against $\prod_j \alpha_j \prod_k \gamma_k$. List the reflected first group in reverse order. Here $Jf = \rho(\Theta_\circ)\bar{f}$, and ϱ is precisely the tensor pullback in (108). The relation $\varrho(-r)\rho(\Theta_\circ) = \rho(\Theta_\circ)\varrho(r)$ fixes the signs in the first group. For small real r, s the density identity gives

$$\langle A, B\rangle_{r,s} = \langle A(r), B(s)\rangle_\circ. \tag{125}$$

For the fixed finite family choose $\delta > 0$ so that rotations with $|r| < \delta$ keep every positive support and every successive displacement within each tuple in $\mathcal{C}_+$. The kernel integral is jointly holomorphic for

$$(\zeta, \xi) \in O^2, \qquad O = \{z : |\Re z| < \delta\}. \tag{126}$$

Indeed, the reflected points lie in $-\mathcal{W}$ and the positive points in $\mathcal{W}$. Independent imaginary rotations preserve these cones and the successive displacements within each tuple. The displacement from the last reflected point to the first unreflected point is a sum of two elements of $\mathcal{W}$. On compact subsets of O^2 the insertion bounds are uniform on the supports, justifying integration. Thus every hypothesis of Lemma 5.1 is satisfied.

At $(\zeta, \xi) = (-i\beta, i\beta)$ the two groups in (124) are the same imaginary rotation of the original reflected configuration. Their profiles are $\varrho(i\beta)Jf$ and $\varrho(i\beta)g$. For real angles near zero the integral agrees with the rotated moment. Holomorphic uniqueness in the vertical strip, together with (117), gives

$$\langle A, B\rangle_\beta := \langle A, B\rangle_{-i\beta,i\beta} = S_{a+b}(J\chi, \eta) + D_{a+b}[J\chi, \eta](i\beta), \qquad \beta \in \mathbb{R}. \tag{127}$$

Here $J\chi$ is the reflected, conjugated tuple. Its concatenation with η is narrow by (120), which is why the finite Fourier expansion (117) applies. The form on the left is positive semidefinite. Empty tuples give the vacuum and are included in this statement. Positivity applies to the continued moment in (127).

8. Eliminating Fourier coefficients

The degree comparison and the use of parity below adapt the arguments of Streater's Lemmas 2 and 3 [Str72, pp. 113–114]; see also Osipov [Osi77, p. 115]. We distinguish the comparison with the law rotated through $\pi/4$, which supplies the divisibility needed here.

We now finish the proof of Theorem C. Let M_N be the largest $|\ell|$ for which c_ℓ is nonzero near some narrow configuration, and set it to zero if all nonzero frequencies vanish there. Then

$$M_0 = 0, \qquad M_N \in 4\mathbb{Z}, \qquad 0 \le M_N \le 3N. \tag{128}$$

The conjugation identity exchanges positive and negative frequencies, so whenever $M_N > 0$ the coefficient at $-M_N$ is nonzero as well. These coefficients are invariant under common translations and permutations of the entries. Translation invariance follows from $R(\theta)^*\tau_a = \tau_{R(-\theta)a}R(\theta)^*$ in both terms of D_N.

*Formal monomials give the natural map $j : \mathcal{X}_{\mathbb{C}} \to L^2_{+,\mathbb{C}}$; no choice of representation of an L^2 observable is needed. All positivity and Cauchy–Schwarz assertions use only finite families.

8.1. $2M_{a+b} \le M_{2a} + M_{2b}$. We apply the leading-degree argument of Streater [Str72, Lemma 2, pp. 113–114] to the positive continued reflection form (127).

Fix nonnegative integers a, b. If $M_{a+b} = 0$, there is nothing to prove, so assume $M_{a+b} > 0$. By Lemma 7.2, there are admissible tuples χ, η of lengths a, b such that

$$c_{-M_{a+b}}[J\chi, \eta] \neq 0. \tag{129}$$

Equation (127) and the finite Fourier expansion (117), applied to this narrow tuple, give

$$\langle A, B\rangle_\beta = \big(c_{-M_{a+b}}[J\chi, \eta] + o(1)\big)e^{M_{a+b}\beta} \qquad (\beta \to +\infty). \tag{130}$$

Similarly, $\langle A, A\rangle_\beta = O(e^{M_{2a}\beta})$ and $\langle B, B\rangle_\beta = O(e^{M_{2b}\beta})$. All three tuples are narrow by the admissibility conditions. The form is positive semidefinite, so Cauchy–Schwarz gives

$$|\langle A, B\rangle_\beta|^2 \le \langle A, A\rangle_\beta \langle B, B\rangle_\beta. \tag{131}$$

Since the leading coefficient in (130) is nonzero, comparison of the exponents yields

$$2M_{a+b} \le M_{2a} + M_{2b}. \tag{132}$$

8.2. $M_{2a} \in 8\mathbb{Z}$. We use the leading-term positivity argument of Streater [Str72, §6, especially the proof of Lemma 8], here for the continued reflection form. Suppose that $M = M_{2a} > 0$. Multiplying the positive form $\langle -, -\rangle_\beta$ on admissible degree-a monomials by $e^{-M\beta}$ and letting $\beta \to +\infty$ gives a positive semidefinite form

$$(A, B) \longmapsto c_{-M}[J\chi, \eta]. \tag{133}$$

The testing lemma supplies A, B for which this form is nonzero. If every diagonal value were zero, Cauchy–Schwarz would make every entry zero. Thus some diagonal value is strictly positive:

$$c_{-M}[J\chi, \chi] > 0. \tag{134}$$

This is a sign assertion about the leading coefficient of $\langle A, A\rangle_\beta$; mixed Fourier coefficients need not have a sign.

The remainder of the argument involves studying the effect of a 45° rotation $Q = R(\pi/4)$ on the measure μ. Define $\tilde{\mu}$ by

$$\tilde{\Phi}(\chi) = \Phi(Q^*\chi), \qquad \tilde{S}_N(\chi) = S_N(Q^*\chi). \tag{135}$$

Since Q permutes the four reflection hyperplanes, reflection invariance and positivity, the regularity estimate and translation invariance are preserved. These are all the hypotheses used in the chosen plane, so the same argument applies to $\tilde{\mu}$, in its own Hilbert space.

Since planar rotations commute,

$$\tilde{D}_N[\chi](\theta) = D_N[\chi](\theta + \pi/4) - D_N[\chi](\pi/4). \tag{136}$$

Matching the nonzero Fourier coefficients gives

$$\tilde{c}_\ell[\chi] = e^{i\ell\pi/4}c_\ell[\chi] = (-1)^{\ell/4}c_\ell[\chi], \qquad \ell \in 4\mathbb{Z} \setminus \{0\}. \tag{137}$$

Nonvanishing is unchanged, so $\tilde{M}_N = M_N$. Now use the positivity of $\tilde{c}_{-M}$ and c_{-M} on the tuple in (134):

$$0 \le \tilde{c}_{-M}[J\chi, \chi] = (-1)^{M/4}\underbrace{c_{-M}[J\chi, \chi]}_{>0}. \tag{138}$$

Thus $(-1)^{M/4} = 1$, which means $M \in 8\mathbb{Z}$. The zero case is automatic, so

$$M_{2a} \in 8\mathbb{Z} \qquad (a \ge 0). \tag{139}$$

It remains to combine the inequality between moment orders, the even-order divisibility and the linear bound. The inequality makes the even-order increments nondecreasing; divisibility prevents a

positive increment from being small enough to respect the bound. The following lemma gives the version needed here.*

Lemma 8.1. Suppose $0 = M_0, M_1, M_2, \dots$ are nonnegative integers satisfying

$$M_n \le 3n, \qquad 2M_{a+b} \le M_{2a} + M_{2b}, \qquad M_{2n} \in 8\mathbb{Z} \tag{140}$$

for all nonnegative integers a, b, n. Then $M_n = 0$ for every n. ■

Proof. Note that $0 \le M_2 \le 6$ and $M_2 \in 8\mathbb{Z}$, so $M_2 = 0$. Define $\delta_n = M_{2n} - M_{2n-2}$. Applying convexity, the second inequality, with $a = n - 1$ and $b = n + 1$ yields

$$2M_{2n} \le M_{2n-2} + M_{2n+2}, \tag{141}$$

hence $\delta_n \le \delta_{n+1}$. The increments are nondecreasing multiples of eight and $\delta_1 = 0$, so all are nonnegative. If δ_r were positive, it would be at least eight, as would every later increment. This implies

$$M_{2j} \ge M_{2r-2} + 8(j - r + 1) \ge 8(j - r + 1), \tag{142}$$

contradicting $M_{2j} \le 6j$ for large j. Therefore every increment vanishes, and $M_{2n} = 0$. Finally, $0 \le 2M_n \le M_0 + M_{2n} = 0$ controls the remaining terms. □

Remark. Convexity (132) makes the even-order increments nondecreasing, and $M_{2n} \in 8\mathbb{Z}$ makes them multiples of eight. Once one is positive, every later increment is at least eight, contradicting $M_{2j} \le 6j$ for sufficiently large j.

9. Conclusion

Lemma 8.1 gives $M_N = 0$ for every N, so (117) gives $D_N[\chi](\theta) = 0$ near every narrow configuration. This proves Theorem C. The narrowing and small-angle arguments give Theorem B; the proof of Theorem A from Theorem B in §2 then gives covariance under rotations of the tx-plane. The same proof applies to every coordinate plane. These rotations generate $\mathrm{SO}(d)$, and the coordinate reflections already preserve the law, so the conclusion holds for all of $\mathrm{O}(d)$.

For compactly supported tests with pairwise disjoint supports this is precisely the distributional vanishing off Δ established above. General Schwartz tuples with pairwise disjoint supports follow by compact truncation, which preserves disjointness, and (13). The theorem concerns these off-diagonal moments; no removal of contact terms has been used.

10. Sharpness of the conclusion and the weight restriction

The hypotheses permit noncovariant contact terms. Moreover, the representation restriction cannot be enlarged uniformly to include planar weight three. Both limitations already occur in dimension two.†

10.1. **Noncovariant contact terms.** Let $\mathcal{T} = \mathrm{Sym}^2_0(\mathbb{R}^2)$, with orthonormal basis

$$e_1 = \frac{dt^2 - dx^2}{\sqrt{2}}, \qquad e_2 = \frac{dt \otimes dx + dx \otimes dt}{\sqrt{2}}.$$

Take independent scalar white noises in these two components, with variances $a, b > 0$, where $a \ne b$. Their characteristic function is

$$\Phi(\chi) = e^{-Q(\chi)/2}, \qquad Q(\chi) = \int_{\mathbb{R}^2} (a\chi_1^2 + b\chi_2^2).$$

The law is translation invariant and satisfies $\mathbb{E}e^{F(\chi)} \le e^{\max(a,b)\|\chi\|_2^2/2}$.

*The proof of Streater's Lemma 3 [Str72, p. 114] likewise combines his convexity inequality with linear growth and parity. Its numerical statement and conclusion differ from those below.

†The two examples in this section were supplied in the referee report.

In this basis the coordinate reflections act by $\mathrm{diag}(1,-1)$ and the diagonal reflections by $\mathrm{diag}(-1,1)$. They preserve Q. If the tests lie on one side of any required mirror, their reflected supports are disjoint from the original supports. Hence

$$\Phi(\chi_j - \Theta^*\chi_k) = e^{-Q(\chi_j)/2} e^{-Q(\chi_k)/2},$$

a rank-one positive semidefinite matrix. Thus all the required reflection positivities hold.

The tensor pullback of $R(\theta)$ is $R(-2\theta)$. At $\theta = \pi/4$ it exchanges e_1, e_2, up to sign. Taking $\chi = f e_1$ shows that $Q(\chi) = a\|f\|_2^2$, whereas $Q(R(\pi/4)^*\chi) = b\|f\|_2^2$. The law is therefore not rotation invariant. Its covariance kernel is $\mathrm{diag}(a,b)\delta(q-q')$. Every nonempty moment on pairwise disjoint supports nevertheless vanishes, since the centered Gaussian evaluations are independent. The failure of covariance is entirely supported at coincidences.

10.2. **Failure at planar weight three.** Let ϕ be the massive scalar Gaussian free field on $\mathbb{R}^2$, with covariance $(-\Delta + m^2)^{-1}$, $m > 0$. This field is reflection positive; see the scalar covariance calculation and its Fock-space extension in [Uhl79, pp. 119–120].

Let $\mathcal{T}_3$ be the two-dimensional space of real harmonic cubics. Choose an orthonormal basis (e_1, e_2) proportional to

$$t^3 - 3tx^2, \qquad 3t^2x - x^3.$$

For the clockwise rotations used here,

$$\rho_3(R(\theta)) = R(-3\theta), \qquad \rho_3(\Theta_\circ) = \mathrm{diag}(-1,1).$$

Thus the planar weights are ± 3. Put $S = \mathrm{diag}(1,-1)$ and define the $\mathcal{T}_3$-valued field by

$$F = (\partial_t\phi, -\partial_x\phi), \qquad F(\chi) = \phi(L\chi), \qquad L\chi = -\operatorname{div}(S\chi).$$

For every symmetry g of the square,

$$S\rho_3(g) = g^{\mathsf{T}} S.$$

Indeed, this holds for a quarter turn and for $\Theta_\circ$, which generate the group; the pullback multiplication order preserves the identity. Since $gS\rho_3(g) = S$, the chain rule gives

$$L(g^*\chi) = (L\chi) \circ g.$$

The map L preserves supports and commutes with translations. Consequently scalar reflection positivity transfers to the characteristic function of F across all four required mirrors. The law is translation invariant.

With unitary Fourier transform,

$$\mathbb{E}F(\chi)^2 = \int_{\mathbb{R}^2} \frac{|p_t\widehat{\chi}_1 - p_x\widehat{\chi}_2|^2}{|p|^2 + m^2}\,\mathrm{d}p \le \|\chi\|_2^2.$$

Thus $\mathbb{E}e^{F(\chi)} \le e^{\|\chi\|_2^2/2}$ for every real test, with the required uniform L^2 bound.

It remains to check that covariance fails away from coincidences. Write $h(r)$ for the scalar covariance at positive separation. The matrix-valued covariance kernel is

$$K(z) = -S(D^2h)(z)S.$$

For two first-component insertions initially separated by $(r,0)$, the pullback rotates the separation to $rn_\theta = R(-\theta)(r,0)$ and the profile to $v_\theta = \rho_3(R(\theta))e_1$. In particular,

$$n_\theta = (\cos\theta, \sin\theta), \qquad Sv_\theta = (\cos 3\theta, -\sin 3\theta), \qquad (Sv_\theta)\cdot n_\theta = \cos 4\theta.$$

The radial Hessian formula therefore yields

$$v_\theta^{\mathsf{T}} K(rn_\theta) v_\theta = -\frac{1}{2}\left(h''(r) + \frac{h'(r)}{r}\right) - \frac{1}{2}\left(h''(r) - \frac{h'(r)}{r}\right)\cos 8\theta.$$

Its difference from the value at $\theta = 0$ is

$$\frac{1}{2}\left(h''(r) - \frac{h'(r)}{r}\right)(1 - \cos 8\theta).$$

The coefficient is strictly positive. Indeed,

$$h(r) = \int_0^\infty \frac{e^{-m^2 s - r^2/(4s)}}{4\pi s}\,\mathrm{d}s, \qquad h''(r) - \frac{h'(r)}{r} = \int_0^\infty \frac{r^2}{4s^2}\frac{e^{-m^2 s - r^2/(4s)}}{4\pi s}\,\mathrm{d}s > 0.$$

For $r > 0$, exponential decay at both ends justifies differentiating under the integral. At $\theta = \pi/8$ the difference is nonzero. Since the kernel is smooth away from the origin, the difference remains nonzero after integration against suitable smooth tests with sufficiently small disjoint supports.

This example has $M_2 = 8$. Allowing weight three would replace the angular bound by $M_N \le 4N$, which permits this value. Hence the cutoff two is sharp for a theorem covering every representation in the stated class: the next integer cutoff already fails in dimension two.

Appendix A. Well-definedness of S_N

Fröhlich [Frö74, Lemma 3.2, pp. 276–277] obtains factorial moment bounds from a uniform strip bound on the generating functional by Cauchy's formula. For the present regularity bound, a tail estimate gives the order dependence used in (13).

For real $\chi \neq 0$, apply (4) to $u\chi$, $u \in \mathbb{R}$. The exponential Markov inequality, optimized over positive and negative u, gives

$$\mu\{|F(\chi)| > C\|\chi\|_1 + r\} \le 2\exp\left(-\frac{r^2}{4C\|\chi\|_2^2}\right), \qquad r > 0. \tag{143}$$

Integrating the tail yields

$$\|F(\chi)\|_{L^p(\mu)} \le K(\|\chi\|_1 + \sqrt{p}\,\|\chi\|_2), \qquad 1 \le p < \infty, \tag{144}$$

with K depending only on C. Consequently $\chi \mapsto F(\chi)$ extends continuously from $\mathcal{S}$ to $(L^1 \cap L^2)(\mathbb{R}^d; \mathcal{T})$, with values in every finite $L^p(\mu)$; the extensions agree. Approximation in both norms and Fatou's lemma extend (4) to these real tests. The complex-linear extension from $(L^1 \cap L^2)(\mathbb{R}^d; \mathcal{T})_{\mathbb{C}}$ to $L^p(\mu)_{\mathbb{C}}$ satisfies (144) by splitting into real and imaginary parts.

Hölder's inequality proves (13), hence joint continuity on Schwartz tests. The Schwartz kernel theorem makes S_N a tempered distribution on $(\mathbb{R}^d)^N$, with one tensor index per variable. The characteristic function is given by the absolutely convergent series

$$\Phi(\chi) = \sum_{N=0}^{\infty} \frac{i^N}{N!} S_N(\chi, \dots, \chi). \tag{145}$$

Indeed, $e^{|F(\chi)|} \le e^{F(\chi)} + e^{-F(\chi)}$ is integrable, so dominated convergence applies to the exponential series.

Invariance of the law under an isometry T is therefore equivalent to invariance of every moment on all tests, not merely disjoint ones. Equality of characteristic functions gives equality of the finite-dimensional evaluation laws, and evaluations generate the Borel sigma-algebra of the weak dual.* This establishes the analytic assertions about moments used in the introduction.

*Choose increasing seminorms p_j defining the Schwartz topology and a countable dense set of tests. Each equicontinuous set $E_{j,k} = \{F : |F(\chi)| \le k p_j(\chi) \text{ for all } \chi\}$ is evaluation-measurable, since its inequalities can be checked on that dense set. On $E_{j,k}$ the weak topology is induced by these countably many evaluations and is second countable. Its Borel sets therefore belong to the restricted evaluation sigma-algebra. The countable union of the $E_{j,k}$ is $\mathcal{S}'$, proving the assertion.

Appendix B. Reflection invariance from reflection positivity and translation invariance

Fix a hyperplane $H \subset \mathbb{R}^d$.

Proposition B.1. Suppose that $\mu : \mathrm{Borel}(\mathcal{S}') \to [0,1]$ is a Borel probability measure whose law is translation-invariant and reflection positive across H. Then the law is invariant under the reflection Θ across H. That is,

$$\Phi(\chi) = \Phi(\Theta^*\chi). \tag{146}$$

■

Proof. Without loss of generality, it suffices to consider $H = \{t = 0\}$. Because Φ is continuous by dominated convergence, and because compactly supported test functions are dense in $\mathcal{S}$, it suffices to consider real compactly supported χ.

First suppose that χ is supported in $\{t > 0\}$. Reflection positivity applied to $\chi, 0$ says that the matrix

$$M = \begin{pmatrix} \Phi(\chi - \Theta^*\chi) & \Phi(\chi) \\ \Phi(-\Theta^*\chi) & 1 \end{pmatrix} \tag{147}$$

is positive semidefinite: $z^\dagger M z \geq 0$ for all $z \in \mathbb{C}^2$. In particular, $z^\dagger M z \in \mathbb{R}$. This can only hold for all z if M is Hermitian, hence

$$\begin{aligned} \Phi(\chi) = \overline{\Phi(-\Theta^*\chi)} &= \int_{\mathcal{S}'} e^{-iF(-\Theta^*\chi)} \, \mathrm{d}\mu(F) \\ &= \int_{\mathcal{S}'} e^{iF(\Theta^*\chi)} \, \mathrm{d}\mu(F) = \Phi(\Theta^*\chi), \end{aligned}$$

as desired. Here $F(\chi) \in \mathbb{R}$ because these tests and distributions are real.

To extend the conclusion to all compactly supported χ, use translation covariance. Let $T_b : (t, x, y) \mapsto (t + b, x, y)$. If $b > 0$ is large enough that $T_{-b}^*\chi$ is supported in $\{t > 0\}$, then

$$\Phi(\chi) = \Phi(T_{-b}^*\chi) = \Phi(\Theta^* T_{-b}^* \chi) = \Phi(T_b^* \Theta^* \chi) = \Phi(\Theta^*\chi). \tag{148}$$

The first and last equalities use translation invariance of the law. □

AI statement

This manuscript is a hybrid of AI and human writing.

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